SRMECOTECH

Software Requirements Specification

GPS Processing Layer: fleet telemetry streaming platform (gps-streaming-service and PulseStream IoT Console)

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This document describes software owned by SRMECOTECH. All IMEIs, registration numbers and company names used as examples are dummy values. Requirements marked “Proposed” or “Future” are not yet approved for implementation.

Document Control

This document is placed under configuration control. Changes are made only through the change-control procedure in the companion document “SRS Rules and Regulations” and are recorded in the revision history below. A version number ending in “.0” denotes a baselined (approved) issue; any other number denotes a working draft.

Table 1: *Revision history*

Version	Date	Author	Description of change
0.1	19 Oct 2026	S. Chattopadhay	Initial requirement notes (README, Hinglish) for the Kafka pipeline, storage and trips.
0.5	28 Oct 2026	S. Chattopadhay	Kafka Streams, data safety, stream consumers, reports and console captured in project guides.
0.9	29 Oct 2026	S. Chattopadhay	First formal SRS per ISO/IEC/IEEE 29148:2018: requirement IDs, priorities, verification methods, business rules, regulations, RTM and open issues.
0.11	3 Nov 2026	S. Chattopadhay	Transport management (three sign-in roles, shipments, portals), plug-in modules, geofencing, speed alert policy, route plans with route deviation, wrong direction and live ETA, and waypoints moved from Future to Baseline (sections TMS, MOD, GEO, SPD, RTE, WPT); GraphHopper fallback engine.

Table 2: *Review and approval*

Role	Name	Signature	Date
Product owner			
Technical lead / architect			
Quality assurance lead			
Information security reviewer			

Table 3: *Distribution list*

Recipient	Purpose
Development team	Design and implementation baseline
Quality assurance	Test design (Section 7 and the RTM in Section 8)
Operations / DevOps	Deployment, monitoring and recovery requirements
Product owner and business stakeholders	Scope confirmation and acceptance

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines what the GPS Processing Layer shall do, how well it shall do it, and the constraints it shall satisfy. It is the agreed reference for design, implementation, verification and acceptance, and it is written so that a developer can implement the system and an independent tester can verify it without having to guess the author's intent.

The document covers two things at once. First, it records the **baseline**: the behaviour of release 0.1.x as implemented in the repository SRMECOTECH/processing-layer (commit c92e25f, 29 October 2026, with transport management, geofencing, the speed alert policy, route deviation, live ETA and waypoints added up to 3 November 2026). Second, it states the **proposed** requirements that are needed before the platform carries real customer traffic. Each requirement is labelled with its status so that the two are never confused (Section 1.6).

1.2 Scope

The product is the **GPS Processing Layer**, a fleet-telemetry streaming platform for goods vehicles. It takes in raw position frames from GPS tracking devices (in this release, a fleet simulator that replays real trips), validates and normalises them, and then stores them. It keeps the live location of every vehicle, detects trips, stops and driving events, and produces day and month reports with fuel-cost estimates. Operators use the PulseStream IoT Console (a web application) to monitor the fleet and the pipeline.

Table 4: *Business objectives*

ID	Objective	Measure of success
OBJ-01	Fleet visibility	The latest valid position of every tracked vehicle can be seen on the live map and through the API.
OBJ-02	Trustworthy data	No reading that the platform has acknowledged is lost, stored twice or silently discarded.
OBJ-03	Operational events	Over-speed, stoppage, parking, idling, ignition and harsh-driving events are detected automatically.
OBJ-04	Trip and stop history	For every vehicle and period, trips and stops with distance and duration can be retrieved.
OBJ-05	Cost reporting	Day and month reports show distance, idling, fuel cost and achievable savings, each with its calculation.
OBJ-06	Operability	Every stage of the pipeline is observable, and failures can be diagnosed and recovered from the console.
OBJ-07	Extensibility	A new processing module can be attached as a new consumer group without changing the existing ones.
OBJ-08	Security and compliance	Access is controlled, secrets are protected and location data is handled lawfully.

1.2.1 In scope

- Ingestion of raw GPS frames (fleet simulator, single-frame test API), with a durable ingest journal in front of Apache Kafka.
- Validation, de-duplication, ordering, distance and movement calculation (the normaliser), and rejection with a reason.
- Storage of positions in PostgreSQL; live location in Redis; master data that resolves IMEI to vehicle and tenant.

- Trip and stop detection; the speed-violation, vehicle-state, stoppage and ignition stream consumers; vehicle events.
- The Kafka Streams normaliser (exactly-once) and analytics application (vehicle table, five-minute activity windows, shadow parity).
- Dead-letter queue and archives of dead letters and rejected frames; the data-safety report.
- Day, month and vehicle-day reports, fuel and savings, cost assumptions, and clearing the database while keeping the reports.
- Run-time switching of external connections (Redis, Valhalla); the engine dashboard and REST API; demo tools.
- The PulseStream IoT Console with sign-in for three roles: GPS provider (DEVELOPER), transporter and consigner, each seeing only its own trucks.
- Transport management: shipment files (the smart_truck tta_trips layout, Excel, CSV), device-to-truck mapping, the transporter and consigner portals; every processor is a plug-in module with its own consumer group.
- Geofencing (grid index, circle and polygon tests, the smart_truck confirmation rules), the day-wise geofence report, the speed alert policy and geofence speed limits.
- Route plans on Valhalla (with GraphHopper as fallback): route deviation, alternate roads, wrong direction, live ETA, destination reached, and waypoints.

1.2.2 Out of scope for this release

- Device protocol parsing (TCP/UDP listeners for vendor protocols). The interface to a future parser is specified (IFR-KAF-001).
- ClickHouse storage and notification delivery (SMS, e-mail, push). The console shows these as “Planned” with example data; their requirements are listed as “Future” for traceability only.
- Map matching of the real trail (Valhalla trace_attributes) and ETA confidence bands (FR-RTE-015, FR-RTE-016, Future).
- Single sign-on, and roles beyond GPS provider, transporter and consigner (FR-AUTH-010).
- Driver management, maintenance scheduling, billing, mobile applications and map-tile hosting.
- Writing to the legacy smart_truck database, which is a read-only source.

1.3 Intended Audience and Reading Suggestions

Table 5: *Audience*

Reader	Recommended sections
Product owner, business stakeholders	1, 2, 3.22 (business rules), 4.10 (regulations), 7.2 (acceptance)
Developers and architects	2, 3, 5, 6 and the companion Database Schema Specification [R4]
Testers / quality assurance	3, 4, 7 and the traceability matrix in Section 8
Operations / DevOps	2.4, 4 (reliability, availability, observability), 5.4, 6.2
Security and compliance reviewers	3.1, 4.5, 4.6, 4.10, 6.2

1.4 Definitions, Acronyms and Abbreviations

Table 6: *Glossary*

Term	Definition
Accepted frame	A raw frame that passed validation and became a processed position.
At-least-once	Delivery guarantee under which a record is never lost but may be delivered more than once.
Changelog topic	Compacted Kafka topic that backs up a Kafka Streams state store so that the store can be restored.
Consumer group	Set of Kafka consumers that share the work of one purpose; each group receives every record of a topic.
Dead letter	A record that a consumer could not handle and parked in topic gps.dead-letters with its error.
Design peak load	150 vehicles replayed at 30 times real time: about 75 frames per second (Section 4.1).
Device assignment	Period during which a device is installed in a vehicle (valid_from to valid_to).
Exactly-once (EOS)	Kafka transactions: output records, state-store updates and input offsets commit together or not at all.
Frame / reading	One position report from a device: IMEI, recorded time, latitude, longitude, speed, optional ignition and odometer.
IMEI	International Mobile Equipment Identity; the identifier of a tracking device.
Ingest journal	Local, fsynced, append-only file store in which frames are written before they are sent to Kafka.
Late / out-of-order reading	A reading whose recorded time is earlier than that of the newest reading already processed for the device.
Market vehicle	A vehicle not owned by any tenant on the platform (owner_tenant_id is null).
Normal operating load	150 vehicles reporting at their real interval (about one frame per minute each): about 2.5 frames per second.
Normaliser	The stage that turns raw frames into processed positions or rejections.
Stale reading	A reading more than 24 hours older than the newest reading of the same device.
Tenant	A customer company; identified on telemetry by its short lower-case code.
ACK	Acknowledgement
API	Application Programming Interface
CO_2	Carbon dioxide
DLQ	Dead-letter queue
DPDP	Digital Personal Data Protection (Act, 2023, India)
EOS	Exactly-once semantics
GEO set	Redis sorted set indexed by geohash for radius queries
HA	High availability
IQ	Kafka Streams Interactive Queries
IST	Indian Standard Time (UTC+05:30)
JSON	JavaScript Object Notation
KRaft	Kafka Raft metadata mode (no ZooKeeper)
KTable	Kafka Streams table abstraction
MoSCoW	Must, Should, Could, Won't (prioritisation)
p95	95th percentile
RBAC	Role-based access control
RPO / RTO	Recovery point objective / recovery time objective
RTM	Requirements Traceability Matrix

Term	Definition
SRS	Software Requirements Specification
TBD	To be determined (recorded in the open-issues register, Section 8.3)
UTC	Coordinated Universal Time

1.5 References

Table 7: *Referenced documents*

Ref.	Document
[R1]	ISO/IEC/IEEE 29148:2018, Systems and software engineering: Life cycle processes, Requirements engineering.
[R2]	IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications (superseded; historical reference only).
[R3]	SRM-GPS-SRS-RUL-001, SRS Rules and Regulations (companion document: how this SRS is written, reviewed and changed).
[R4]	SRM-GPS-DBS-001, Database Schema Specification for gps_tracking (companion document).
[R5]	Repository README.md and README_SIMPLE.md (project overview, design decisions).
[R6]	docs/DATA_SAFETY_GUIDE.md (no-loss design: journal, retention, dead letters, data-safety report).
[R7]	docs/KAFKA_STREAMS_GUIDE.md (normaliser and analytics topologies).
[R8]	docs/REPORTS_GUIDE.md (report formulas, summary job, clear database).
[R9]	docs/ENGINE_AND_DOCKER_GUIDE.md and docs/APPLICATIONS_GUIDE.md (operations).
[R10]	Apache Kafka 4.3 documentation: Kafka Streams architecture, transactions and consumer groups.
[R11]	Traccar open-source GPS tracking server: motion, distance, filter, ignition and engine-hours algorithms ported in this product.
[R12]	The Digital Personal Data Protection Act, 2023 (India), and the rules made under it.
[R13]	The Information Technology Act, 2000 (India), section 43A, and the IT (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011.
[R14]	CERT-In Directions under section 70B(6) of the IT Act, 28 April 2022 (incident reporting and log retention).
[R15]	AIS-140, Intelligent Transportation Systems: requirements for public transport vehicle operation (vehicle location tracking devices), ARAI.
[R16]	W3C Web Content Accessibility Guidelines (WCAG) 2.1.

1.6 Document Conventions

1.6.1 Requirement language

- **shall**: a mandatory requirement. Every “shall” statement is verifiable and carries a unique identifier.
- **should**: a recommendation. Not meeting it needs a recorded justification.
- **may**: a permitted option.
- **will**: a statement of fact or intent about something outside the system (for example, an external system's behaviour). It is not a requirement on this product.

1.6.2 Requirement identifiers

Table 8: Identifier scheme

Prefix	Meaning	Example
FR-<area>-nnn	Functional requirement	FR-VAL-003 (validation), FR-TRP-002 (trip detection)
NFR-<cat>-nnn	Non-functional requirement	NFR-PERF-001, NFR-REL-004, NFR-SEC-002
IFR-<if>-nnn	Interface requirement	IFR-API-001 (REST), IFR-KAF-001 (Kafka), IFR-RDS-001 (Redis)
DR-nnn	Data requirement	DR-004 (time zone of stored timestamps)
BR-nnn	Business rule	BR-001 (one device in one vehicle at a time)
REG-nnn	Regulatory or legal requirement	REG-001 (personal-data notice and purpose)
CON-nnn	Design or implementation constraint	CON-004 (no third-party UI libraries)
ASM-nnn / DEP-nnn	Assumption / dependency	ASM-002, DEP-001
OBJ-nn	Business objective (source of requirements)	OBJ-02 (trustworthy data)
AC-nn / TC-<req>	Acceptance criterion / test case	AC-03; TC-FR-VAL-003 verifies FR-VAL-003

Identifiers are never re-used. A deleted requirement keeps its row with status “Deleted” and the reason.

1.6.3 Priority, verification method and status

Table 9: Attribute values used in requirement tables

Attribute	Values
Priority	Must: required for acceptance of the release. Should: important, may move to the next release with stakeholder agreement. Could: desirable.
Verify	T = Test (executed test with expected result). I = Inspection (review of code, configuration or documents). A = Analysis (calculation, model or log analysis). D = Demonstration (observed operation without measurement).
Status	Baseline: implemented in release 0.1.x; must be verified before approval. Proposed: not implemented; needs approval. Future: roadmap item kept for traceability; not part of this release's acceptance.

1.6.4 Units, time and configurable values

- Distances are in metres (m) or kilometres (km), speeds in km/h, durations in seconds (s) unless stated otherwise.
- Timestamps are instants in UTC, written in ISO-8601 (for example 2026-10-29T05:17:11Z). Report days use Asia/Kolkata (IST).
- Many thresholds can be configured. A configuration key is shown in code font with its default, for example `gps.trips.min-distance-meters` (default 200). Such a requirement applies to the **configured** value; the default is the value that is verified at acceptance.

2 Overall Description

2.1 Product Perspective

The GPS Processing Layer is a new, self-contained system that sits between the tracking devices and the applications that use fleet data. It replaces ad-hoc processing on the legacy smart_truck MySQL database, which it reads (read-only) only as a source of real trips for the simulator. Figure 1 shows the system boundary and the external entities.

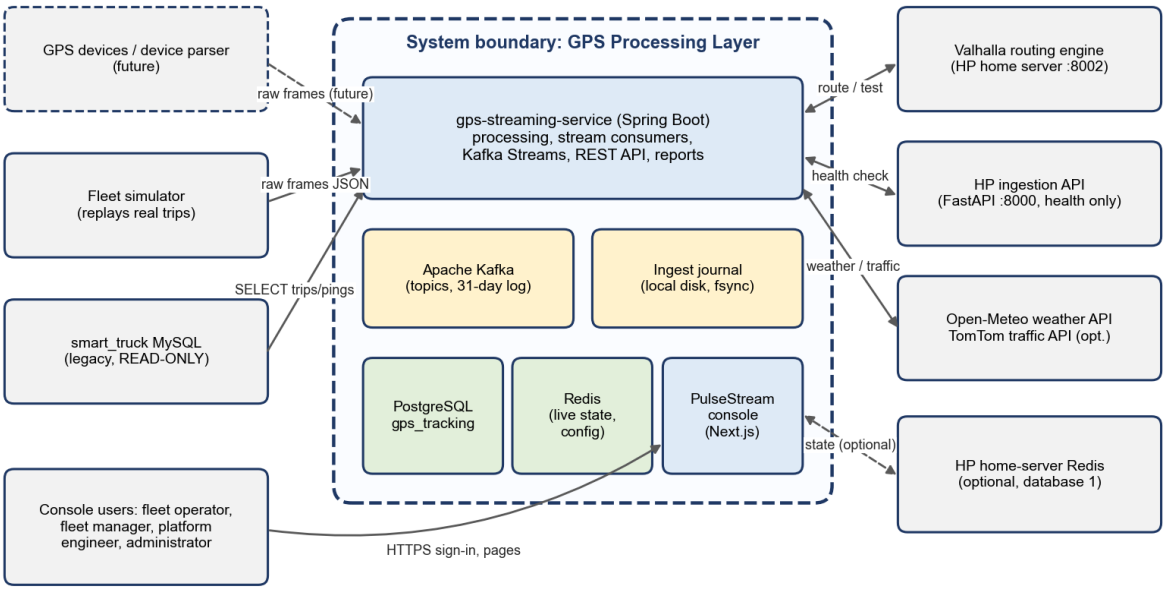


Figure 1: *System context diagram*

Inside the boundary, one Spring Boot service (gps-streaming-service) hosts the processing logic, the stream consumers, two Kafka Streams applications, the REST API and the report jobs. Apache Kafka is the buffer and the 31-day record of all traffic. PostgreSQL (database gps_tracking) is the system of record. Redis holds live state and the speed rules. The PulseStream IoT Console is a separate Next.js application that proxies its /api calls to the service.

Table 10: *Software modules*

Module	Responsibility
gps-configuration	Names of topics, consumer groups, stages and Redis keys; message records; JSON mapping. Plain Java.
gps-connectivity	All connections to the outside: Kafka publishers and listeners, retry and dead letters, ingest journal, Redis stores, JPA entities and repositories, smart_truck reader, Valhalla client, metrics.
gps-processing	Business logic as pure functions: validation, movement and distance, trip detection, speed violation, vehicle state (Kalman filter), stoppage, ignition, fuel estimation. Fully unit-tested.
gps-streams	Kafka Streams topologies: the exactly-once normaliser and the analytics application (vehicle table, activity windows, parity).
gps-application	The Spring Boot service: Kafka and direct flows, simulator, master data, storage, live location, trips, events, reports, connections, data safety, REST API, Flyway migrations.
gps-dashboard-ui	PulseStream IoT Console (Next.js, React). Sign-in, seven sections (Live, Processing, Alerts, Storage, Analytics, Notifications, Developer), Reports.

2.2 Product Functions

Figure 2 shows how a frame travels through the platform in the Kafka flow (the default). The functions are summarised in Table 11 and specified in Section 3.

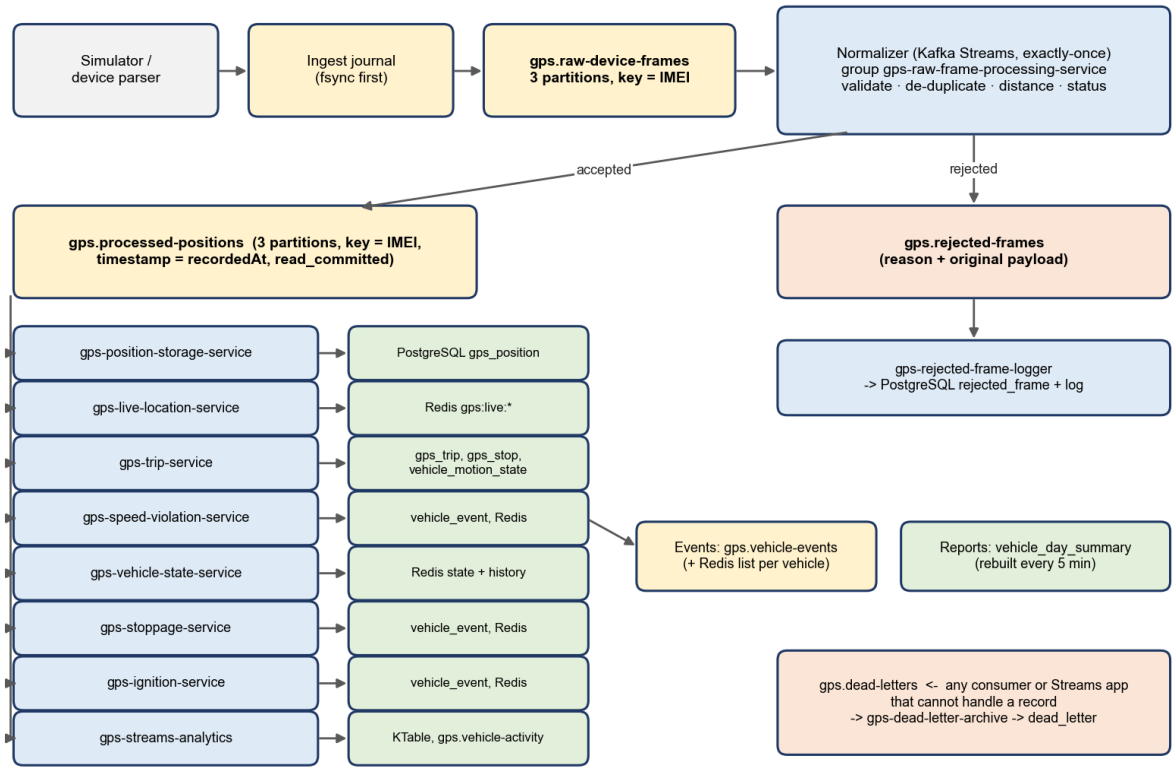


Figure 2: *Processing pipeline (Kafka flow)*

Table 11: *Summary of product functions*

Function	Summary	Section
Access control	Sign-in to the console; signed session cookie; lock-out after repeated failures.	3.1
Master data	Tenants, vehicles, devices and time-bounded device assignments; IMEI resolution.	3.2
Ingestion	Accept frames, journal them durably, publish them to Kafka keyed by IMEI.	3.3
Validation and normalisation	Reject malformed frames with a reason; de-duplicate; compute distance and movement.	3.4, 3.5
Storage and live location	Store each position once; keep the newest position of each vehicle in Redis.	3.6, 3.7
Trips and stops	Detect trips and stops as readings arrive.	3.8
Stream consumers and events	Over-speed, sensor mismatch, vehicle state, stoppage, parking, ignition, idling, harsh driving.	3.9 to 3.13
Kafka Streams	Exactly-once normaliser; vehicle table; activity windows; shadow parity.	3.14
No-data-loss tooling	Dead letters and rejected-frame archives; replay and discard; data-safety report.	3.15, 3.16
Reports	Day, month, vehicle-day, fuel and savings, cost assumptions, clear database.	3.17

Function	Summary	Section
Administration	Connections, simulator, engine dashboard, demo reset, console navigation.	3.18 to 3.21

2.3 User Classes and Characteristics

Table 12: *User classes*

User class	Description and goals	Frequency / expertise	Main functions
Fleet operator	Watches vehicles and alerts during the shift; reacts to stoppages and over-speed.	Daily, continuous / non-technical	Live, Alerts
Fleet manager	Reviews day and month reports, fuel cost and savings; sets cost assumptions and speed limits.	Daily to monthly / business user	Reports, Alerts, Processing rules
Platform engineer	Develops and operates the pipeline; investigates lag, rejects and dead letters.	Daily / expert	Processing, Storage, Developer
Administrator	Manages connections, simulator, data clearing and credentials.	Occasional / expert	Developer, Reports → Clear database
Demo presenter	Runs demonstrations with the simulator and resets the data between them.	Occasional / intermediate	Developer → Demo tools

> **Note:** In this release all user classes share one console account (FR-AUTH-002). Separating them by role is proposed in FR-AUTH-010.

2.4 Operating Environment

Table 13: *Operating environment (baseline)*

Component	Version	Notes
Java runtime	21	Service runs on a JRE in a non-root container (uid 1001).
Spring Boot	4.1.1	With Spring for Apache Kafka, Spring Data JPA (Hibernate), Flyway, Actuator.
Apache Kafka	4.3.1 (KRaft)	One broker by default; three brokers with docker-compose.ha.yml.
Kafka Streams / RocksDB	4.2.1	Library inside the service; state directory on a volume.
PostgreSQL	14 (14.22 native)	Database gps_tracking; extension btree_gist.
Redis	7.4 (Docker, Redis Stack) or 8.2 (native)	Database 0 locally; database 1 on the HP home server.
MySQL (smart_truck)	8	Legacy, read-only source for the simulator.
Node.js / Next.js / React	Next.js with React (see gps-dashboard-ui/package.json)	Console; no third-party UI packages (CON-004).
Docker Desktop / Compose	29.x	Recommended deployment (containers postgres, kafka, redis, gps-service, dashboard).
Host operating system	Windows 11 (development), any Docker host	Native Windows scripts in infrastructure/windows.
Client browser	Current Chrome, Edge or Firefox	Desktop screens 1280 px wide or more (NFR-CMP-002).

2.5 Design and Implementation Constraints

Table 14: *Constraints*

ID	Requirement	Priority	Verify	Status
CON-001	The schema of database gps_tracking shall be created and changed only by numbered Flyway migrations; Hibernate shall run with `ddl-auto: validate`, and an applied migration shall never be edited.	Must	I	Baseline
CON-002	The legacy smart_truck database shall be accessed through a separate connection pool opened read-only, using SELECT statements only.	Must	I	Baseline
CON-003	Passwords and keys shall not be stored in application.yml or in version control; they shall come from local-secrets.properties or .env (both git-ignored) or environment variables.	Must	I	Baseline
CON-004	The console shall use no third-party UI packages: only next, react and react-dom, CSS Modules and in-house icons (client server policy).	Must	I	Baseline
CON-005	Every Kafka topic of the platform shall be keyed by IMEI and have 3 partitions; the name of each topic, consumer group and Redis key shall be defined in exactly one class (KafkaTopicNames, KafkaConsumerGroupNames, RedisKeyNames).	Must	I	Baseline
CON-006	Processing rules shall be implemented as plain Java without Spring, Kafka or database dependencies, so that the Kafka flow, the direct flow and Kafka Streams run the same code.	Must	I	Baseline
CON-007	Normalisation shall have exactly one owner at a time (`gps.streams.normalization` = owner, shadow or off); the Kafka Streams normaliser shall use the listener's group id so that ownership can change without replay or gap.	Must	I	Baseline
CON-008	The service shall start even when smart_truck, Valhalla, the HP ingestion API or the weather and traffic APIs are unreachable; only the features that need them shall fail.	Must	T	Baseline

2.6 Assumptions and Dependencies

Table 15: *Assumptions*

ID	Assumption	Impact if false
ASM-001	Device clocks are synchronised to GNSS time within 5 minutes.	Frames are rejected as “in the future”; trips are skewed.
ASM-002	Each device reports at most one reading per second, and a vehicle's readings arrive mostly in order.	More late readings; trip distances need the repair job (OI-07).
ASM-003	The fleet in scope is about 150 goods vehicles, fitted with speed governors (observed maximum 60 km/h).	Capacity requirements (4.2) must be re-validated.
ASM-004	A device is moved between vehicles only by recording a new device assignment.	Readings are attributed to the wrong vehicle.
ASM-005	Operators use the console from a trusted network (laptop, LAN or VPN) in this release.	Proposed security requirements NFR-SEC-006 to 008 become mandatory.

Table 16: *Dependencies*

ID	Dependency	Used for
DEP-001	Apache Kafka cluster (1 or 3 brokers), 31-day retention	Transport and 31-day replay
DEP-002	PostgreSQL 14 with btree_gist	System of record
DEP-003	Redis 7.4+ (local or HP home server, database 1)	Live state, rules, history
DEP-004	smart_truck MySQL 8 (read-only user)	Simulator source trips

ID	Dependency	Used for
DEP-005	Valhalla routing engine (HP home server, all-India map, truck costing)	Route plans and re-plan checks (FR-RTE)
DEP-008	GraphHopper 11.1 with the India graph (this laptop, compose profile routing)	Fallback routing engine (FR-RTE-002)
DEP-006	Open-Meteo weather API; TomTom traffic API (optional key)	Stoppage cause analysis
DEP-007	HP ingestion API (FastAPI), health endpoint only	Connections page status

3 Specific Requirements

This section states the functional requirements module by module. Each module gives its purpose, inputs and outputs, followed by its requirements, including the behaviour for invalid input, duplicates, late data, missing permissions and unavailable dependencies. Section 3.29 summarises the exceptional cases across modules.

3.1 User Authentication and Session Management (AUTH)

Inputs: user name, password, “Keep me signed in”. Outputs: signed session cookie `pulsestream_session`; redirect to the requested page. Component: gps-dashboard-ui (session.ts, auth-actions.ts, proxy.ts).

Table 17: *Authentication requirements*

ID	Requirement	Priority	Verify	Status
FR-AUTH-001	The console shall serve no page and forward no /api/* request without a valid session. An unauthenticated page request shall be redirected to /login, with the requested path kept and restored after sign-in. An unauthenticated /api/* request shall receive HTTP 401.	Must	T	Baseline
FR-AUTH-002	The console shall accept exactly one account, whose user name and password are read from DASHBOARD_USERNAME and DASHBOARD_PASSWORD.	Must	T	Baseline
FR-AUTH-003	When DASHBOARD_USERNAME or DASHBOARD_PASSWORD is not set, the console shall use the development account “abc” / “abc” (see NFR-SEC-006).	Must	T	Baseline
FR-AUTH-004	Credential comparison shall take the same time whichever part (user name or password) is wrong.	Should	I	Baseline
FR-AUTH-005	After 5 failed sign-in attempts from the same client address, the console shall refuse further attempts from that address for 60 seconds and say so.	Must	T	Baseline
FR-AUTH-006	On success, the console shall issue a session cookie signed with AUTH_SECRET (or, if unset, a key derived from the account with PBKDF2), valid for 12 hours, or 30 days when “Keep me signed in” is selected.	Must	T	Baseline
FR-AUTH-007	A missing, altered or expired session cookie shall be treated as no session.	Must	T	Baseline
FR-AUTH-008	Sign-out shall delete the session cookie and return the user to the sign-in page.	Must	T	Baseline
FR-AUTH-009	Changing the account password or AUTH_SECRET shall invalidate every existing session.	Should	T	Baseline
FR-AUTH-010	The system should support named user accounts with the roles Viewer, Operator and Administrator, and restrict every state-changing operation (speed rules, connections, cost assumptions, dead-letter replay or discard, simulator, clear database) to Operator or Administrator.	Should	T	Proposed

3.2 Master Data and Device Resolution (MD)

Inputs: device IMEI and the reading's recorded time. Outputs: tenant code and vehicle registration number valid at that time. Component: DeviceVehicleDirectory, SimulationMasterDataSeeder, TenantCodeFormatter; tables tenant, vehicle, device, device_assignment.

Table 18: *Master data requirements*

ID	Requirement	Priority	Verify	Status
FR-MD-001	The system shall store tenants, vehicles, devices and device assignments as specified in the Database Schema Specification [R4].	Must	I	Baseline
FR-MD-002	The system shall resolve a reading to the vehicle and tenant of the device assignment whose period [valid_from, valid_to) contains the reading's recorded time, not the processing time.	Must	T	Baseline
FR-MD-003	When no assignment covers the reading's time, the system shall still store the position, with tenant and vehicle empty, and log a WARN line naming the IMEI.	Must	T	Baseline
FR-MD-004	The system shall cache each resolved assignment period and query the database again only when a reading falls outside the cached period.	Should	T	Baseline
FR-MD-005	The simulator shall register the tenants, vehicles, devices and assignments of the devices it replays before sending their first frame.	Must	T	Baseline
FR-MD-006	A tenant code shall be derived from the company name by lower-casing it, replacing every run of characters other than a–z and 0–9 with one hyphen, trimming hyphens at both ends and cutting it to 40 characters (for example “DEMO TRANSPORT, JSR” → “demo-transport-jsr”).	Must	T	Baseline
FR-MD-007	The system should provide API and console functions to create, edit, activate, suspend and close tenants, register vehicles and devices, and open and close device assignments, with validation of BR-001 to BR-003.	Should	T	Proposed

3.3 GPS Frame Ingestion (ING)

Inputs: raw frame JSON (Table 62) from the simulator, the single-frame API, or a future device parser. Outputs: a record in gps.raw-device-frames. Component: RawGpsFrameSender, KafkaRawGpsFrameSender, IngestJournal, DurableKafkaSender.

Table 19: *Ingestion requirements*

ID	Requirement	Priority	Verify	Status
FR-ING-001	The system shall publish every raw frame to topic gps.raw-device-frames with the frame's IMEI as the record key, so that all frames of a device are kept in order in one partition.	Must	T	Baseline
FR-ING-002	Before sending a frame to Kafka, the system shall append it to the ingest journal and fsync it to disk (`gps.journal.fsync-every-record` = true).	Must	T	Baseline
FR-ING-003	A journal entry shall be removed only after Kafka has acknowledged the record with acks=all.	Must	T	Baseline
FR-ING-004	When Kafka is unavailable, the sender shall return without blocking the caller for more than 5 seconds (`max.block.ms`), keep the frame in the journal, and resend journalled frames in their original order when Kafka is available again.	Must	T	Baseline
FR-ING-005	After a crash or restart, the system shall resend every journalled frame that Kafka had not acknowledged, in its original order.	Must	T	Baseline
FR-ING-006	The system shall provide POST /api/simulator/send-single-frame that accepts one raw frame (valid or invalid) and processes it like any other frame.	Must	T	Baseline
FR-ING-007	In the direct flow (`gps.processing-flow=direct`), frames shall be processed synchronously in arrival order, one at a time, without Kafka, the journal or the dead-letter queue.	Should	T	Baseline

ID	Requirement	Priority	Verify	Status
FR-ING-008	A device parser (TCP/UDP listener for vendor protocols, AIS-140 included) shall convert device messages into raw frames and use the same sender.	Could	T	Future

3.4 Frame Validation (VAL)

Each rule is applied in the order listed. The first failing rule rejects the frame with the stated reason, and the frame is published to gps.rejected-frames with its original payload. Component: GpsFrameValidationRules, ValidateRawFrameStep.

Table 20: *Validation rules*

ID	Requirement	Priority	Verify	Status
FR-VAL-001	A frame that is not valid JSON shall be rejected with reason “Frame is not valid JSON”.	Must	T	Baseline
FR-VAL-002	A frame with a missing or blank IMEI shall be rejected with reason “IMEI is missing”.	Must	T	Baseline
FR-VAL-003	A frame without a recorded time shall be rejected with reason “Recording time is missing”.	Must	T	Baseline
FR-VAL-004	A frame whose recorded time is more than 5 minutes after the processing time shall be rejected with reason “Recording time <t> is in the future”.	Must	T	Baseline
FR-VAL-005	A frame without latitude or longitude shall be rejected with reason “Latitude or longitude is missing”.	Must	T	Baseline
FR-VAL-006	A frame with latitude outside −90 to 90 shall be rejected with reason “Latitude <v> is outside -90..90”.	Must	T	Baseline
FR-VAL-007	A frame with longitude outside −180 to 180 shall be rejected with reason “Longitude <v> is outside -180..180”.	Must	T	Baseline
FR-VAL-008	A frame at exactly 0, 0 shall be rejected with reason “Position is 0,0 - the device has no GPS fix”.	Must	T	Baseline
FR-VAL-009	A frame without speed shall be rejected with reason “Speed is missing”; a missing value shall never be read as 0.	Must	T	Baseline
FR-VAL-010	A frame with speed outside 0 to 150 km/h shall be rejected with reason “Speed <v> km/h is outside 0..150”.	Must	T	Baseline
FR-VAL-011	The Kafka Streams normaliser shall reject a frame whose record key differs from the frame's IMEI.	Must	T	Baseline

3.5 Normalisation, Distance and Movement (NRM)

Inputs: a valid frame and the previous reading of the device. Outputs: a processed position (Table 63). Component: CalculateMovementStep, DistanceCalculator, MovementRules, NormalizeFrameProcessor.

Table 21: *Normalisation requirements*

ID	Requirement	Priority	Verify	Status
FR-NRM-001	A reading with the same IMEI and recorded time as a reading seen in the last 24 hours (`gps.streams.seen-readings-retention`) shall be rejected with reason “Duplicate reading”.	Must	T	Baseline
FR-NRM-002	A reading more than 24 hours older than the device's newest reading (`gps.streams.max-lateness`) shall be rejected with reason “Stale reading”.	Must	T	Baseline

ID	Requirement	Priority	Verify	Status
FR-NRM-003	A reading older than the device's newest reading (and not stale) shall be accepted with arrived_out_of_order = true, distance 0, seconds since previous 0 and distance source OUT_OF_ORDER, and shall not become the device's newest reading.	Must	T	Baseline
FR-NRM-004	For an in-order reading, distance shall be the device odometer delta when the device reports a believable odometer (source DEVICE_ODOMETER), otherwise the great-circle distance from the previous position (source GPS).	Must	T	Baseline
FR-NRM-005	When the position moved less than 10 m (`gps.distance.jitter-meters`) and the speed is below 5 km/h (`jitter-max-speed-kmph`), the position shall be clamped to the previous one and the distance shall be 0 (source JITTER_FILTERED).	Must	T	Baseline
FR-NRM-006	When the implied speed exceeds 200 km/h (`max-implied-speed-kmph`) over a jump longer than 500 m (`min-jump-meters`), the distance shall not be counted (source JUMP_FILTERED) and the new position shall be kept.	Must	T	Baseline
FR-NRM-007	The first reading of a device shall have distance 0 and source FIRST_READING.	Must	T	Baseline
FR-NRM-008	total_distance_m shall equal the previous total plus the counted distance of the reading.	Must	T	Baseline
FR-NRM-009	A reading shall be MOVING when its speed is at least 5 km/h or it is at least 50 m from the previous reading; otherwise STOPPED.	Must	T	Baseline
FR-NRM-010	The previous reading shall be remembered only after the result was delivered; a reading whose result could not be delivered shall be processed normally on retry and not be treated as a duplicate.	Must	T	Baseline
FR-NRM-011	After a restart, the previous reading of a device shall be recovered from the state store (Kafka Streams) or from PostgreSQL (listener), so that the first reading after a restart has the correct distance.	Must	T	Baseline
FR-NRM-012	The normaliser shall set the output record timestamp to the reading's recorded time.	Must	T	Baseline

3.6 Position Storage (STO)

Table 22: *Position storage requirements*

ID	Requirement	Priority	Verify	Status
FR-STO-001	Consumer group gps-position-storage-service shall store every processed position in table gps_position with the tenant code and vehicle registration number resolved per FR-MD-002.	Must	T	Baseline
FR-STO-002	A position whose (imei, recorded_at) already exists shall not be stored again, and the delivery shall be treated as successful.	Must	T	Baseline
FR-STO-003	The system shall provide GET /api/gps-positions/count, GET /api/gps-positions/recently-stored (newest 20) and GET /api/vehicles/{registrationNo}/latest-position.	Must	T	Baseline

3.7 Live Location (LIVE)

Table 23: *Live location requirements*

ID	Requirement	Priority	Verify	Status
FR-LIVE-001	Consumer group gps-live-location-service shall keep, for each IMEI, the newest position in Redis hash `gps:live:vehicle:{imei}` (vehicle, tenant, latitude, longitude, speed, status, recordedAt, updatedAt) and in GEO set `gps:live:positions`.	Must	T	Baseline
FR-LIVE-002	A reading older than the stored live position shall not overwrite it.	Must	T	Baseline
FR-LIVE-003	The console shall show the live map and vehicle list from the live positions, refreshing every 2 seconds (default polling interval) while live updates are on.	Must	D	Baseline

3.8 Trip and Stop Detection (TRP)

Consumer group gps-trip-service. Inputs: processed positions. Outputs: rows in gps_trip, gps_stop, gps_trip_late_reading and vehicle_motion_state. Figure 3 shows the state machine.

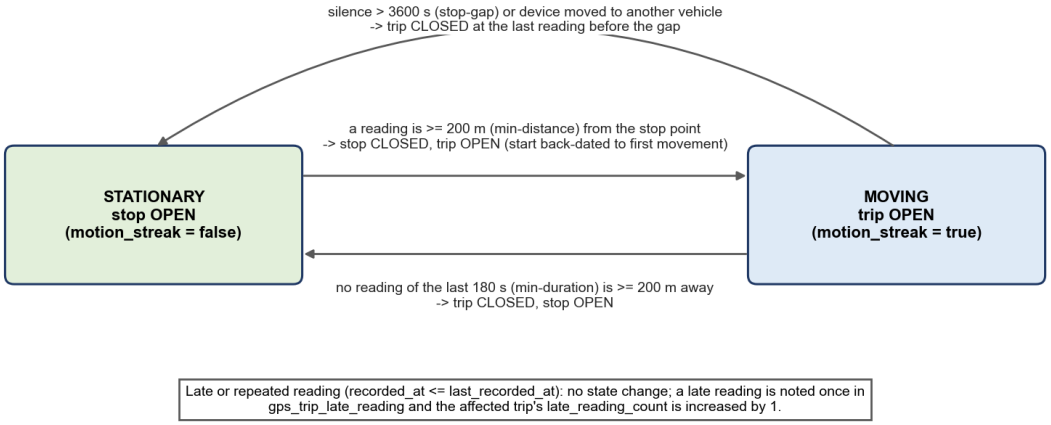


Figure 3: *Trip and stop detection state machine*

Table 24: *Trip and stop requirements*

ID	Requirement	Priority	Verify	Status
FR-TRP-001	When a stationary vehicle has a reading at least 200 m (`gps.trips.min-distance-meters`) from the stop position, the system shall close the stop and open a trip whose start is back-dated to the reading at which the vehicle began to move.	Must	T	Baseline
FR-TRP-002	When none of the readings of the last 180 s (`gps.trips.min-duration-seconds`) is 200 m or more from the newest reading, the system shall close the trip and open a stop.	Must	T	Baseline
FR-TRP-003	When a device is silent for more than 3600 s (`gps.trips.stop-gap-seconds`), the open trip shall be closed at the last reading before the silence.	Must	T	Baseline
FR-TRP-004	When the device assignment of a reading differs from the open trip's assignment, the open trip shall be closed; a trip shall never span two vehicles.	Must	T	Baseline
FR-TRP-005	Trip distance shall be the sum of distance_from_previous_m of the trip's in-order readings and shall never be negative.	Must	T	Baseline

ID	Requirement	Priority	Verify	Status
FR-TRP-006	An open trip or stop shall be stored with status OPEN and updated with each reading; it shall be CLOSED when it ends.	Must	T	Baseline
FR-TRP-007	A late reading (recorded at or before the last applied reading) shall not change state, distance or duration; it shall be recorded once in gps_trip_late_reading and increase the affected trip's late_reading_count by 1.	Must	T	Baseline
FR-TRP-008	The state update and the trip and stop writes of one reading shall be one database transaction, guarded by “last_recorded_at < the reading's recorded time”; a redelivered reading shall change nothing.	Must	T	Baseline
FR-TRP-009	Trips and stops shall be upserted on (imei, start_recorded_at).	Must	T	Baseline
FR-TRP-010	The system shall provide GET /api/trips, GET /api/stops (filters imei, from, to; default the last 24 hours; overlap semantics) and GET /api/trips/count.	Must	T	Baseline
FR-TRP-011	A repair job should recompute the trips listed in gps_trip_late_reading from gps_position.	Could	T	Future

3.9 Speed Violation (SPD)

Consumer group gps-speed-violation-service. Rules are stored in Redis (`gps:config:speed-rule:{scope}`). The default rule is: limit 60 km/h, 3 consecutive readings, 60 s minimum duration, 5 km/h hysteresis, tolerance max(10 km/h, 15 %), 180 s maximum gap, 3-reading GPS window, 3 mismatching readings.

Table 25: *Speed violation requirements*

ID	Requirement	Priority	Verify	Status
FR-SPD-001	The rule applied to a reading shall be the vehicle rule (scope vehicle:{registration}) if present, else the tenant rule (tenant:{code}), else the default rule.	Must	T	Baseline
FR-SPD-002	An over-speed episode shall start when at least minConsecutivePings consecutive readings exceed the limit and the excess has lasted at least minDurationSeconds; an OVERSPEED event shall then be raised once with status OPEN.	Must	T	Baseline
FR-SPD-003	An episode shall end, and its event become CLOSED, only when the speed falls below the limit minus the hysteresis, or when the gap between readings exceeds maxGapSeconds.	Must	T	Baseline
FR-SPD-004	For each reading, the system shall compare the reported speed with the GPS speed over the last windowPings readings (path ÷ time, or odometer when available) and classify it GPS_VERIFIED or SENSOR_SUSPECT using the tolerance max(toleranceKmph, tolerancePercent × speed).	Must	T	Baseline
FR-SPD-005	After mismatchPings consecutive SENSOR_SUSPECT readings, the system shall raise a SPEED_SENSOR_MISMATCH event.	Must	T	Baseline
FR-SPD-006	The system shall provide GET /api/speed/rules, PUT /api/speed/rules/{scope}, DELETE /api/speed/rules/{scope} (the default rule cannot be deleted) and GET /api/speed/active, and create the default rule on first start.	Must	T	Baseline

3.10 Vehicle State and History (VST)

Table 26: *Vehicle state requirements*

ID	Requirement	Priority	Verify	Status
FR-VST-001	Consumer group gps-vehicle-state-service shall keep per vehicle in `gps:state:vehicle:{imei}`: heading, acceleration, smoothed (Kalman-filtered) position and distance, data-quality counters and moving and standing time.	Must	T	Baseline
FR-VST-002	The Kalman filter shall use a constant-velocity model (GPS sigma 12 m, acceleration sigma 1.5), reject outliers by a Mahalanobis gate, and reset after 3 consecutive rejections.	Must	T	Baseline
FR-VST-003	HARSH_ACCELERATION and HARSH_BRAKING events shall be computed only from consecutive readings less than 10 s apart.	Must	T	Baseline
FR-VST-004	The history of each vehicle shall be appended to Redis stream `gps:history:{imei}`, trimmed to about 1000 entries (`gps.stream.history-length`).	Must	T	Baseline
FR-VST-005	GET /api/vehicles/{imei}/state shall return all Redis keys of the vehicle, its history (`history` = number of entries), its events, the processing memory and the trip-detector state; GET /api/vehicles/states shall return every vehicle's state.	Must	T	Baseline

3.11 Stoppage and Parking (STP)

Table 27: *Stoppage requirements*

ID	Requirement	Priority	Verify	Status
FR-STP-001	Consumer group gps-stoppage-service shall detect a stop when the vehicle stays within 100 m (`gps.stream.stoppage.radius-meters`) for at least 180 s, with reported speed at most 3 km/h and a low distance-over-time speed; the stop start shall be back-dated to arrival and its position shall be the centroid.	Must	T	Baseline
FR-STP-002	A stop shall end only after 2 consecutive readings outside the radius (`exit-confirm-readings`); a silence longer than 1800 s shall end it; stops less than 180 s apart at the same place shall be merged.	Must	T	Baseline
FR-STP-003	A stop with ignition ON shall raise a STOPPAGE event; a stop with ignition OFF shall raise a PARKED event without alarm. If the ignition changes during the stop, the event kind shall change accordingly.	Must	T	Baseline
FR-STP-004	For each STOPPAGE the system shall run a traffic agent (fleet vehicles with engine on within 1 km, mostly slow, mean a jam; TomTom when a key is set) and a weather agent (Open-Meteo, 0.1° cell, cached 10 minutes), and set the verdict JUSTIFIED_TRAFFIC, JUSTIFIED_WEATHER or UNEXPLAINED (severity HIGH).	Must	T	Baseline
FR-STP-005	If an agent's API does not answer within 3000 ms, the verdict shall be computed without it and the stoppage shall still be recorded.	Must	T	Baseline

3.12 Ignition, Engine Hours and Idling (IGN)

Table 28: *Ignition requirements*

ID	Requirement	Priority	Verify	Status
FR-IGN-001	Consumer group gps-ignition-service shall use the device's ignition input when present, otherwise a virtual ignition: ON while moving, OFF after 300 s standing (`virtual-off-after-seconds`), debounced over 2 readings and back-dated.	Must	T	Baseline

ID	Requirement	Priority	Verify	Status
FR-IGN-002	The system shall raise IGNITION_ON and IGNITION_OFF events and accumulate engine hours, without crediting engine time across long silences.	Must	T	Baseline
FR-IGN-003	The system shall raise an IDLING event when the engine is ON, speed is at most 3 km/h and the vehicle has been standing for more than 300 s.	Must	T	Baseline

3.13 Vehicle Events (EVT)

Table 29: *Vehicle event requirements*

ID	Requirement	Priority	Verify	Status
FR-EVT-001	Every event shall have an IMEI, vehicle, tenant, type (Table 65), severity (INFO, LOW, MEDIUM, HIGH, CRITICAL), status (OPEN, CLOSED), start and end time, position, detecting consumer and details.	Must	I	Baseline
FR-EVT-002	An episode shall be written when it starts (OPEN) and again when it ends (CLOSED) under the same event_key, as an upsert, so that a redelivery has no effect.	Must	T	Baseline
FR-EVT-003	Events shall be written to table vehicle_event, to the vehicle's Redis list `gps:events:{imei}`, and to topic gps.vehicle-events (through the ingest journal).	Must	T	Baseline
FR-EVT-004	The system shall provide GET /api/events (filters type, imei, status, limit ≤ 500) and GET /api/events/summary (per type: last 24 h and currently open).	Must	T	Baseline
FR-EVT-005	Notification channels (SMS, e-mail, push), routing rules, escalations and quiet hours shall deliver events to people.	Could	T	Future

3.14 Kafka Streams Normaliser and Analytics (KS)

Table 30: *Kafka Streams requirements*

ID	Requirement	Priority	Verify	Status
FR-KS-001	With `gps.streams.normalization=owner` (default), the Kafka Streams application gps-raw-frame-processing-service shall normalise raw frames with processing guarantee exactly_once_v2, and the listener for the same group shall not be created.	Must	T	Baseline
FR-KS-002	In shadow mode, the listener shall normalise, Kafka Streams shall write the same results to gps.shadow.processed-positions and gps.shadow.rejected-frames, and GET /api/streams/parity shall compare both outputs reading by reading with verdicts MATCH, MISMATCH (with field), WARMUP, ONLY_LISTENER or ONLY_STREAMS.	Must	T	Baseline
FR-KS-003	The analytics application gps-streams-analytics shall keep a table of each vehicle's latest state and five-minute event-time activity windows with a two-minute grace period, emit one final record per closed window to gps.vehicle-activity, and keep closed windows queryable for 6 hours.	Must	T	Baseline
FR-KS-004	The system shall provide GET /api/streams/overview, /vehicles, /vehicles/{imei}, /fleet/activity?windows=n and /parity, and forward a query to the instance that owns the vehicle's partition (Interactive Queries).	Must	T	Baseline
FR-KS-005	When a Kafka Streams task fails because a dependency (for example the database fallback) is unavailable, the record shall be retried by a replacement stream thread and shall not be dead-lettered.	Must	T	Baseline

ID	Requirement	Priority	Verify	Status
FR-KS-006	Health component `kafkaStreams` shall report DOWN when an application is in state ERROR (all its stream threads have died); REBALANCING shall count as UP. Interactive Queries shall answer HTTP 503 while an application is neither RUNNING nor REBALANCING.	Should	T	Baseline

3.15 Dead Letters and Rejected-Frame Archive (DLQ)

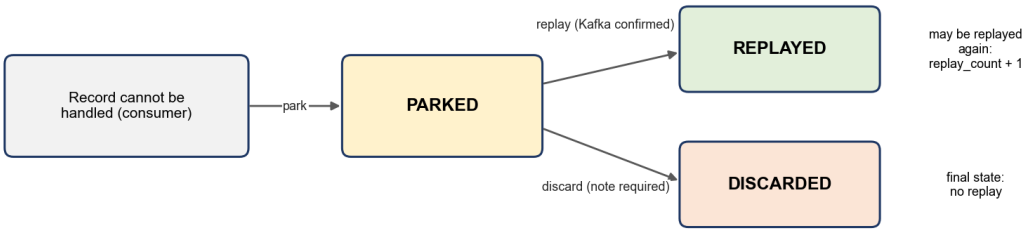


Figure 4: *Dead-letter lifecycle*

Table 31: *Dead-letter requirements*

ID	Requirement	Priority	Verify	Status
FR-DLQ-001	A record that cannot be parsed, or that violates a database integrity rule, shall be parked in gps.dead-letters at once, with headers giving the original topic, partition, offset, consumer group and exception.	Must	T	Baseline
FR-DLQ-002	A failure caused by an unavailable dependency (PostgreSQL, Redis, Kafka, network) shall be retried with back-off 1 s, 2 s, 4 s … capped at 60 s, without limit, and shall never be dead-lettered.	Must	T	Baseline
FR-DLQ-003	Any other failure shall be tried 3 times and then dead-lettered.	Must	T	Baseline
FR-DLQ-004	Consumer group gps-dead-letter-archive shall store every dead letter once in table dead_letter with status PARKED.	Must	T	Baseline
FR-DLQ-005	Replay (POST /api/dead-letters/{id}/replay, or /replay?topic= for the oldest parked) shall send the record to its original topic and set status REPLAYED and replay_count + 1 only after Kafka confirms; otherwise the row shall stay PARKED. A DISCARDED row shall not be replayed.	Must	T	Baseline
FR-DLQ-006	Discard (POST /api/dead-letters/{id}/discard) shall require a non-blank note, keep it in resolution_note and set status DISCARDED; only PARKED rows can be discarded.	Must	T	Baseline
FR-DLQ-007	Consumer group gps-rejected-frame-logger shall store every rejected frame once in table rejected_frame (reason, original payload, payload hash) and log it with its reason.	Must	T	Baseline
FR-DLQ-008	Rows of dead_letter and rejected_frame shall never be deleted, except by the demo reset (FR-ADM-002).	Must	I	Baseline

3.16 Data-Safety Report (SAFE)

Table 32: *Data-safety requirements*

ID	Requirement	Priority	Verify	Status
FR-SAFE-001	GET /api/data-safety shall report, with status OK, WARN or FAIL and an explanation, these checks: retention (a month of data in Kafka), no-loss (no consumer group behind the log start), stopped groups, replication, broker fsync, offsets retention, topic auto-creation disabled, producer acks, ingest journal backlog, and parked dead letters.	Must	T	Baseline
FR-SAFE-002	When a consumer group's committed offset is below the log-start offset of any partition (data deleted before it was read), the no-loss check shall be FAIL.	Must	T	Baseline
FR-SAFE-003	Health component `dataSafety` shall report DOWN while any check is FAIL.	Must	T	Baseline

3.17 Reports (RPT)

Reports read the summary table vehicle_day_summary (one row per vehicle per day), never raw positions, except the vehicle-day sheet of one vehicle. Fuel (litres) = distance ÷ mileage + idling hours × idle burn + (km above the economical speed ÷ mileage) × penalty %. Savings = (idling fuel + extra fuel for fast driving) × price per litre. CO_2 = litres × kg CO_2 per litre.

Table 33: *Report requirements*

ID	Requirement	Priority	Verify	Status
FR-RPT-001	A report day shall run from 00:00 to 24:00 Asia/Kolkata (`gps.reports.zone`).	Must	T	Baseline
FR-RPT-002	The summary job shall rebuild today's and yesterday's rows every 5 minutes (`gps.reports.refresh-every`), summarise up to 62 earlier days that have positions once after start (`backfill-days`), and rebuild on POST /api/reports/refresh.	Must	T	Baseline
FR-RPT-003	A silence longer than 600 s between moving readings, or 3600 s between standing readings, shall count neither as driving nor as standing time.	Must	T	Baseline
FR-RPT-004	GET /api/reports/day?date&tenant&imei shall return every vehicle's day (distance, driving and standing time, stops, idling, maximum speed, over-speed and harsh events, fuel, cost, savings) and fleet totals.	Must	T	Baseline
FR-RPT-005	GET /api/reports/month?month&tenant&imei shall return each vehicle's month and the fleet per day.	Must	T	Baseline
FR-RPT-006	GET /api/reports/vehicle-day shall return one vehicle's day in time order: every trip (km, speed) and stop (duration, place, engine on or off, idling).	Must	T	Baseline
FR-RPT-007	Every figure shown in a report shall have a “How is this calculated?” section giving the formula with that figure's own values.	Must	D	Baseline
FR-RPT-008	Cost assumptions (fuel type, price per litre, mileage, idle burn, economical speed, fast-driving penalty, CO_2 factor) shall be read and changed through GET and PUT /api/reports/assumptions and stored in report_setting; prices shall be applied when a report is read, so that a change re-prices past reports immediately.	Must	T	Baseline
FR-RPT-009	PUT /api/reports/assumptions shall reject a negative price, rate, percentage or CO_2 factor, a mileage of 0 or less, and an economical speed outside 10 to 150 km/h, with a message.	Must	T	Baseline

ID	Requirement	Priority	Verify	Status
FR-RPT-010	Clear database (POST /api/admin/reset?keepReports=true) shall first freeze the live summary rows into a new generation greater than 0 and then empty the raw tables; reports shall add up all generations of a day. With keepReports=false the summaries shall be emptied too.	Must	T	Baseline
FR-RPT-011	Day and month reports shall be printable to PDF from the browser and downloadable as CSV.	Should	D	Baseline

3.18 External Connections Management (CON)

Table 34: *Connections requirements*

ID	Requirement	Priority	Verify	Status
FR-CON-001	GET /api/connections shall show the Redis in use, the Valhalla URL and the HP ingestion API with reachability.	Must	T	Baseline
FR-CON-002	POST /api/connections/redis/test shall report ping, memory, eviction policy and the keys present in the selected database; PUT /api/connections/redis shall switch the Redis in use without a restart and copy the configuration keys (`gps:config:*`).	Must	T	Baseline
FR-CON-003	A switch to a Redis database that contains another program's keys (for example `gps:points`, `devices`) shall be refused with an explanation.	Must	T	Baseline
FR-CON-004	The chosen Redis and Valhalla URLs shall be stored in connection_setting and win over application.yml defaults at the next start.	Must	T	Baseline
FR-CON-005	POST /api/connections/valhalla/test shall compute a short sample route and show it; POST /api/connections/ingestion-api/test shall show the API's health.	Should	T	Baseline
FR-CON-006	Redis calls shall time out after 5 s (connect 3 s) and HTTP calls after 5 s, so that an unreachable server never blocks the console.	Must	T	Baseline

3.19 Fleet Simulator (SIM)

Table 35: *Simulator requirements*

ID	Requirement	Priority	Verify	Status
FR-SIM-001	The simulator shall read the latest trip of up to 150 distinct devices (`vehicle-count`) that has at least 200 pings from smart_truck, and replay up to 500 pings per vehicle.	Must	T	Baseline
FR-SIM-002	In faithful timing, each ping shall keep its real interval; the replay shall start 120 minutes in the past and run 30 times faster until it reaches the present, then in real time.	Must	T	Baseline
FR-SIM-003	At the end of a trip a vehicle shall drive the trip back in reverse, repeating for ever when `repeat-forever` is true; a restarted fleet shall resume after the newest stored reading.	Must	T	Baseline
FR-SIM-004	The simulator shall give an ignition input to 50 % of devices, an odometer to 50 %, and make 25 % of drivers keep the engine running while standing.	Should	T	Baseline
FR-SIM-005	The system shall provide POST /api/simulator/start (vehicleCount, pingsPerVehicle), POST /api/simulator/stop and GET /api/simulator/status, and show a failure reason (for example Kafka unavailable or access denied) in the status.	Must	T	Baseline

3.20 Monitoring, Engine Dashboard and Administration (ADM)

Table 36: *Monitoring and administration requirements*

ID	Requirement	Priority	Verify	Status
FR-ADM-001	The engine dashboard (port 8080) and GET /api/engine/overview shall show, refreshed every 2 s: each producer's and consumer's messages per second (10-second average), 60-second history, totals and failures; each topic partition's offsets; each consumer group's state, partition assignment and lag.	Must	T	Baseline
FR-ADM-002	Demo reset (POST /api/admin/reset) shall stop the fleet, pause consumers, delete the platform topics' records, truncate the telemetry and archive tables (keeping master data, speed rules and report settings), delete only the platform's telemetry keys in Redis, reset the Kafka Streams applications, resume consumers and optionally restart the fleet.	Must	T	Baseline
FR-ADM-003	Demo reset shall be refused when `gps.admin.reset-enabled` is false.	Must	T	Baseline
FR-ADM-004	Every log line written while a record is handled shall carry the stage, IMEI, partition and offset; an ENGINE summary line shall be written every 30 s.	Must	I	Baseline
FR-ADM-005	GET /api/engine/devices?imei= shall filter devices by IMEI fragment, and GET /api/engine/devices/{imei} shall show the device's partition, per-stage counts and live position.	Should	T	Baseline

3.21 Console Navigation and Presentation (UI)

Table 37: *Console requirements*

ID	Requirement	Priority	Verify	Status
FR-UI-001	The console shall offer the sections Live, Processing, Alerts, Storage, Analytics, Notifications and Developer in a top bar, plus Reports, each with its own left menu.	Must	D	Baseline
FR-UI-002	Each card shall state its data source as “Live” (from the service) or “Example data” (module not built); a module that is not built shall be marked “Planned” in the menu.	Must	I	Baseline
FR-UI-003	The console shall offer a command palette (Ctrl+K), light and dark appearance, five accent colours, density, and a switch to pause live updates; preferences shall persist in the browser.	Should	D	Baseline
FR-UI-004	When the service is unreachable, each page shall show an error state naming the failed call instead of stale or empty data.	Must	T	Proposed

3.22 Transport Management and Portals (TMS)

Inputs: shipment files from transporters (the smart_truck tta_trips layout, Excel or CSV), device-to-truck mappings agreed between the GPS provider and the transporter, sign-in with a role. Outputs: shipments followed by GPS, the transporter and consigner portals. Components: tms/ShipmentImportService, ShipmentStore, PortalService, PortalController, auth/AuthController, PortalTokens; dashboard /transporter and /consigner.

Table 38: *Transport management requirements*

ID	Requirement	Priority	Verify	Status
FR-TMS-001	The sign-in page shall ask for the role (GPS provider, transporter, consigner) as well as the user id and password; tokens shall be signed with the key in auth_key and checked on every /api/portal/** request (HTTP 401 without a valid token).	Must	T	Baseline
FR-TMS-002	A transporter or consigner user shall see only the shipments, trucks, alerts, geofence reports and trips of its own company (BR-014).	Must	T	Baseline
FR-TMS-003	The system shall import shipment files in the tta_trips column layout, Excel and CSV, show a preview with the column mapping and problems before committing, and record each upload in shipment_import.	Must	T	Baseline
FR-TMS-004	Uploading the same trip again shall update its shipment (BR-013); an unchanged row shall be skipped by its row hash.	Must	T	Baseline
FR-TMS-005	Origin and destination names in a file shall be resolved to places (geofences) by their normalised code; unknown names shall be listed, not invented.	Must	T	Baseline
FR-TMS-006	Module shipment-tracking shall set departure (cleared the origin fence), arrival (entered the destination fence) and delay (past the ETA) from the earliest reading that proves each, and raise SHIPMENT_DEPARTED, SHIPMENT_ARRIVED and SHIPMENT_DELAYED.	Must	T	Baseline
FR-TMS-007	The GPS provider shall map a device to a truck for a period (device_assignment); a reading shall belong to the truck mapped at its time (BR-004).	Must	T	Baseline
FR-TMS-008	For demonstrations, the GPS provider shall be able to import the shipments behind the trips the fleet simulator replays (POST /api/portal/admin/demo-shipments), through the same importer.	Could	T	Baseline

3.23 Plug-in Processing Modules (MOD)

Every processor after Kafka is a module of gps-module-api: a GpsModule bean with a descriptor (id, consumer group, order, events, Redis state). Components: gps-module-api, modules/ModuleRunner, ModuleKafkaListeners, ModuleReplayService.

Table 39: *Module requirements*

ID	Requirement	Priority	Verify	Status
FR-MOD-001	Each module shall read gps.processed-positions with its own consumer group, keyed by IMEI, so one vehicle's readings reach it in order and a slow module never holds another back.	Must	T	Baseline
FR-MOD-002	A module shall be switched off by one line (gps.modules.{id}.enabled = false) without changing any other module.	Must	T	Baseline
FR-MOD-003	A module shall keep its per-device state in Redis (gps:state:{id}:{imei}) and write its events before its state, so a retried reading finds the same events (event key upsert).	Must	T	Baseline
FR-MOD-004	POST /api/modules/{id}/replay?from= shall stop that module's consumers, clear its state, let it clean its own tables from that moment, move its offsets back and start it again; no other module shall reread.	Must	T	Baseline
FR-MOD-005	GET /api/modules shall list every module with its consumer group, load, lag, failures and whether it runs.	Must	T	Baseline

3.24 Geofencing (GEO)

Module geofence (consumer group gps-geofence-service). Inputs: processed positions; the place table (circles and polygons). Outputs: geofence_visit, GEOFENCE_* events, the live board, the day-wise geofence report. Components: gps-processing geofence/ (GeofenceIndex, Geofence, GeofenceTracker), geofence/GeofenceModule, GeofenceDirectory, GeofenceReportService.

Table 40: *Geofencing requirements*

ID	Requirement	Priority	Verify	Status
FR-GEO-001	Every active geofence shall be filed in every grid cell (0.05 degrees, or the median fence diameter if larger) its bounding box touches; a reading shall be tested only against the fences of its own cell.	Must	T	Baseline
FR-GEO-002	The index shall be rebuilt at once after an edit through the API and within 30 s after any other change to place.	Must	T	Baseline
FR-GEO-003	Inside a circle shall be decided by distance from the centre (equirectangular, cos of the centre latitude); inside a polygon by ray casting (an odd number of border crossings).	Must	T	Baseline
FR-GEO-004	A side of a border shall be believed only after the truck stayed on it 15 minutes, except an exit beyond twice the radius, which counts at once (the smart_truck rules).	Must	T	Baseline
FR-GEO-005	Entry shall be stamped at the first reading inside and exit at the last reading inside, with the gap to the reading outside kept as the uncertainty.	Must	T	Baseline
FR-GEO-006	In and out within the confirmation time while moving at least 300 m shall be a pass-through; a fence crossed between two readings (gap up to 30 min, move at least 200 m) shall be an inferred pass.	Must	T	Baseline
FR-GEO-007	Inside longer than the fence's dwell limit (else its role's: origin 720, destination 360, yard 240, other 360 minutes) shall raise GEOFENCE_DWELL, OPEN until the truck leaves.	Must	T	Baseline
FR-GEO-008	The day-wise report shall give, for one day and the user's trucks, the time per fence against that fence's 30-day usual, hourly crowding, each truck's 24-hour timeline and plain-language insights; an open visit shall end at the device's last reading, not the clock.	Must	T	Baseline
FR-GEO-009	Measured on all 2,220 smart_truck trips (3.48 million readings), the module should find at least 95 % of smart_truck's own geofence events at under 1 microsecond per reading.	Should	T	Baseline

3.25 Speed Alert Policy and Zone Limits (SPD)

The over-speed episodes of FR-SPD-001 to 006 are graded by an alert policy kept in Redis (gps:config:speed-policy). Components: gps-processing speed/SpeedPolicy, streaming/SpeedPolicyStore, SpeedRuleController.

Table 41: *Speed alert policy requirements*

ID	Requirement	Priority	Verify	Status
FR-SPD-007	An over-speed shall be graded WARNING, BREACH (default 10 km/h over) or SEVERE (default 25 km/h over); an episode the GPS contradicted (SENSOR_SUSPECT) shall be lowered to a device-health ticket when the policy says so.	Must	T	Baseline
FR-SPD-008	Inside a geofence with a speed limit, the strictest such limit shall apply when it is lower than the vehicle's rule; the event shall name the zone.	Must	T	Baseline
FR-SPD-009	The policy may make limits stricter at night (default 22:00 to 05:00); GET and PUT /api/speed/policy shall read and change it, refusing inconsistent bands with HTTP 400.	Should	T	Baseline

3.26 Route Plans, Route Deviation and Live ETA (RTE)

Module route-deviation (consumer group gps-route-deviation-service). Inputs: processed positions; route plans; the routing engines. Outputs: route_plan live columns, ROUTE_DEVIATION, ROUTE_ALTERNATE, WRONG_DIRECTION, ETA_CHANGED, DESTINATION_REACHED, the live ETA strips and trip maps. Components: gps-processing route/ (RoutePlan, RouteTracker), route/RoutingEngines, RoutePlanService, RoutePlanDirectory, RouteDeviationModule, RouteLiveService; docs/ROUTE_DEVIATION_ETA_DESIGN.md.

Table 42: *Route deviation and ETA requirements*

ID	Requirement	Priority	Verify	Status
FR-RTE-001	Every open shipment with an origin and a destination geofence shall get a road route plan within 60 s (truck costing), with the geofences its line crosses as stops; a plan may also be made by hand from a start, stops and a destination, each a geofence or a latitude, longitude in India.	Must	T	Baseline
FR-RTE-002	Plans shall come from Valhalla at the address set on Developer → Connections; when it does not answer, from GraphHopper (gps.routing.graphhopper-url).	Must	T	Baseline
FR-RTE-003	A shipment the engines cannot plan shall be tried again after 10 minutes; a manual plan request shall then fail with HTTP 502 naming each engine's problem.	Must	T	Baseline
FR-RTE-004	Each reading of a truck with a plan shall be projected onto the plan's line (segment grid of 0.01 degree cells) to give its distance from the line and its progress along it; a reading of a truck without a plan shall cost one map lookup and write nothing.	Must	T	Baseline
FR-RTE-005	A reading implying more than 150 km/h from the previous one shall be ignored; nothing shall be judged while the truck is inside its origin or a stop.	Must	T	Baseline
FR-RTE-006	Outside the corridor (150 m where the plan is highway speed, 80 m elsewhere) shall only make the truck SUSPECT; back inside for 2 readings shall clear it with no event. A truck seen at its origin that leaves it outside the corridor shall depart as SUSPECT; a truck never seen at the start of its plan shall wait.	Must	T	Baseline
FR-RTE-007	After 3 readings AND 5 minutes AND 1 km driven off the line, or at once beyond 2 km, the system shall ask the engine for the route from the truck's position (with its heading) to the destination, without holding the Kafka consumer.	Must	T	Baseline
FR-RTE-008	If that route, plus the time already driven off the line, is within 1.10 times (or 15 minutes of) the planned remaining time, the system shall raise ROUTE_ALTERNATE and make it the plan (version + 1); otherwise ROUTE_DEVIATION OPEN with the added time, HIGH above 30 and CRITICAL above 90 minutes. If no engine answers, the deviation shall open unverified (MEDIUM).	Must	T	Baseline
FR-RTE-009	Three readings back inside the corridor shall close ROUTE_DEVIATION with minutes off route, kilometres off the plan and extra kilometres.	Must	T	Baseline
FR-RTE-010	WRONG_DIRECTION shall open when progress falls 1 km over 3 moving readings, or the heading is more than 135 degrees against the road for 3 readings with at least 300 m lost, or (off the line) the re-plan grows 2 km longer; it shall close after 3 readings going forward.	Must	T	Baseline

ID	Requirement	Priority	Verify	Status
FR-RTE-011	ETA shall be the reading time plus the planned driving time still ahead × the pace, plus planned stays ahead, plus 30 minutes of rest per 4.5 hours of driving ahead, plus the added time of an open deviation. The pace shall weigh the truck's live pace (moving time ÷ planned time, averaged over 60 minutes, clamped 0.6 to 2.5) against the plan's starting pace (the lane's median of 3+ arrived trips, else 1.0) by moving time since departure, one hour of driving weighing as much as the starting pace.	Must	T	Baseline
FR-RTE-012	The shown ETA shall move at most 2 minutes per minute; it shall jump only when a deviation opens, changes or closes. ETA_CHANGED shall be raised when it moved 30 minutes, or 10 % of the time left if that is more, from the last announced value, with the reason (pace, stopped, deviation).	Must	T	Baseline
FR-RTE-013	A reading inside the destination geofence (or within 300 m of a destination point) shall raise DESTINATION_REACHED with the promised ETA (the ETA at departure) against the actual arrival, and end the plan.	Must	T	Baseline
FR-RTE-014	The console and both portals shall show each trip as a strip (origin, stops, destination, the truck at its progress, coloured by state, ETA against the promise) and on an SVG map with the planned line, the real trail on and off it, the stops and the events; no map library shall be used.	Must	T	Baseline
FR-RTE-015	Map matching (Valhalla trace_attributes) should confirm a suspected deviation before the re-plan check.	Could	T	Future
FR-RTE-016	The ETA should be shown with a confidence band from the lane's past ETA errors.	Could	T	Future

3.27 Waypoints (WPT)

Module waypoints (consumer group gps-waypoint-service). Inputs: processed positions; the stops of the truck's plan. Outputs: route_stop status, WAYPOINT_ARRIVED, WAYPOINT_DEPARTED, WAYPOINT_MISSED. Components: gps-processing route/WaypointTracker, route/WaypointModule.

Table 43: *Waypoint requirements*

ID	Requirement	Priority	Verify	Status
FR-WPT-001	Each stop shall have its place along the plan (chainage), so its order and ETA are known; a stop the current line does not pass shall be marked so.	Must	T	Baseline
FR-WPT-002	Two minutes inside a stop shall be an arrival stamped at the first reading inside; the first reading outside shall be the departure, stamped at the last reading inside, with the stay against the planned stay.	Must	T	Baseline
FR-WPT-003	In and out within two minutes shall be recorded as a drive-through without an event.	Must	T	Baseline
FR-WPT-004	A stop never entered shall be missed once the truck is 2 km past it along the plan, or at the destination; a missed planned stop shall raise WAYPOINT_MISSED (HIGH); a geofence on the way that was not entered shall only be marked missed on the stop.	Must	T	Baseline
FR-WPT-005	An arrival while an earlier planned stop is still ahead shall be marked out of order.	Must	T	Baseline
FR-WPT-006	Each stop ahead shall show its planned time and live ETA (planned driving to it × the trip's pace + the stays before it); a reached stop its arrival against the plan.	Must	T	Baseline

3.28 Business Rules

Business rules are policies of the business that the software enforces. They apply in every module.

Table 44: *Business rules*

ID	Requirement	Priority	Verify	Status
BR-001	A device shall be installed in at most one vehicle at any instant (enforced by an exclusion constraint on device_assignment).	Must	T	Baseline
BR-002	A vehicle registration number shall be unique across the platform; a vehicle without an owner tenant is a market vehicle.	Must	T	Baseline
BR-003	A tenant code shall match ^[a-z0-9][a-z0-9_-]*$, be at most 40 characters and never be reused; tenant status is ACTIVE, SUSPENDED or CLOSED.	Must	T	Baseline
BR-004	A reading belongs to the vehicle and tenant assigned at the reading's recorded time.	Must	T	Baseline
BR-005	A device has at most one reading per instant: (imei, recorded_at) is unique.	Must	T	Baseline
BR-006	Nothing received is discarded silently: each frame ends as a stored position, an archived rejection, or an archived dead letter.	Must	A	Baseline
BR-007	A goods vehicle's reported speed above 150 km/h is a device error, not an over-speed.	Must	T	Baseline
BR-008	Speed-limit precedence is vehicle, then tenant, then the platform default.	Must	T	Baseline
BR-009	A stop with the engine off is parking and raises no alarm; a stop with the engine on is a stoppage and is analysed for its cause.	Must	T	Baseline
BR-010	Over-speed events are operational information for the fleet owner; the system shall not present them as legal traffic offences (speed limits differ by road, state and vehicle class).	Must	I	Baseline
BR-011	Trip distance comes only from in-order readings and is never negative; late readings are counted but never change a trip's distance.	Must	T	Baseline
BR-012	Configuration (speed rules, cost assumptions, connections) and master data survive Clear database and the demo reset.	Must	T	Baseline
BR-013	A shipment is identified by its transporter and trip number: uploading the same trip again updates it, never duplicates it.	Must	T	Baseline
BR-014	A transporter or consigner user sees only its own trucks and shipments; the GPS provider sees everything.	Must	T	Baseline
BR-015	The legacy smart_truck database and project are read-only references: the platform copies from them and never writes to them.	Must	I	Baseline
BR-016	A truck follows at most one active route plan; a new plan closes the previous one.	Must	T	Baseline
BR-017	Leaving the planned road is a deviation only when the way taken is clearly slower than the plan; another road about as good is an alternate, not an alarm.	Must	T	Baseline

3.29 Exceptional Conditions Summary

The table below collects how the system behaves in exceptional conditions, with the requirements that govern each case.

Table 45: *Exceptional conditions and required behaviour*

Condition	Required behaviour	Requirements
Invalid input (bad JSON, missing or out-of-range field)	Reject with a reason, archive in rejected_frame, continue with the next frame.	FR-VAL-*, FR-DLQ-007
Duplicate delivery from Kafka	Rejected by the normaliser or ignored by idempotent writes; no second row.	FR-NRM-001, FR-STO-002, FR-TRP-008, FR-EVT-002
Out-of-order reading	Accepted for history with zero distance; never becomes the newest; does not change trips or live position.	FR-NRM-003, FR-LIVE-002, FR-TRP-007
Reading older than 24 hours	Rejected as stale with its payload.	FR-NRM-002
Device clock in the future	Rejected when more than 5 minutes ahead.	FR-VAL-004
Kafka down	Frames stay in the journal; the caller is not blocked; sent in order later.	FR-ING-004
PostgreSQL or Redis down	Consumers retry with back-off without limit; records wait in Kafka.	FR-DLQ-002
Unprocessable record (bug)	Three tries, then parked as a dead letter; later records are not blocked.	FR-DLQ-003
Service crash or restart	Uncommitted records are delivered again; journal resent; state restored.	FR-ING-005, FR-NRM-011, NFR-REL-003
Consumer behind Kafka retention	Data-safety report FAIL; health DOWN; loss is never hidden.	FR-SAFE-002, FR-SAFE-003
Device unassigned at reading time	Position stored without vehicle and tenant; WARN log.	FR-MD-003
Device silent for a long time	Trip closed at last reading; stop or episode ended.	FR-TRP-003, FR-SPD-003, FR-STP-002
External API slow or down (weather, traffic, Valhalla)	Time out; continue without it.	FR-STP-005, FR-CON-006, CON-008
No permission (no session)	Redirect to sign-in, or HTTP 401 for /api.	FR-AUTH-001
Repeated wrong passwords	Lock the client address for 60 s.	FR-AUTH-005
Two operators change the same rule	Last write wins; the change is logged (proposed audit).	NFR-SEC-009 (Proposed)
Capacity limit reached	Lag grows but nothing is lost while within 31 days; data-safety report warns.	NFR-CAP-002, FR-SAFE-001
Routing engine down when planning	GraphHopper is tried after Valhalla; if neither answers, the shipment is tried again 10 minutes later; a manual plan gets HTTP 502.	FR-RTE-002, FR-RTE-003
Routing engine down during a suspected deviation	The deviation is opened as unverified (MEDIUM) instead of waiting.	FR-RTE-008
GPS jump far off the road	A reading implying more than 150 km/h is ignored by the route tracker.	FR-RTE-005
Fence edited while trucks are inside	Index rebuilt within 30 s; a removed fence ends its open visits (FENCE_REMOVED).	FR-GEO-002, FR-GEO-007

4 Non-Functional Requirements

Unless stated otherwise, measurements are made on the reference deployment (docker-compose.yml on one host with at least 4 CPU cores, 16 GB RAM and an SSD) at the load defined below. Thresholds marked “Proposed” are engineering targets to be confirmed with stakeholders after a baseline test (OI-01).

Table 46: *Load definitions*

Item	Value
Normal operating load	150 vehicles in faithful timing, after catch-up: about 2.5 frames per second.
Design peak load	150 vehicles during catch-up (30 times real time): about 75 frames per second.
Measurement interval	15 minutes of steady load after a 5-minute warm-up, unless stated otherwise.

4.1 Performance

Table 47: *Performance requirements*

ID	Requirement	Priority	Verify	Status
NFR-PERF-001	At design peak load, the lag of every consumer group, sampled every second, shall be at most 225 records (3 seconds of input) at the 95th percentile over the measurement interval, with zero failures.	Must	T	Proposed
NFR-PERF-002	At design peak load, for 95 % of stored positions, stored_at − processed_at shall be at most 1 second.	Must	T	Proposed
NFR-PERF-003	At design peak load, for 95 % of frames, the vehicle's live position in Redis shall reflect the frame within 2 seconds of the frame being processed.	Must	T	Proposed
NFR-PERF-004	GET /api/reports/day for 150 vehicles shall respond within 2 s and GET /api/reports/month within 3 s at the 95th percentile, with the summary rows already built.	Must	T	Proposed
NFR-PERF-005	Every REST query endpoint other than reports shall respond within 1 s at the 95th percentile at normal operating load.	Should	T	Proposed
NFR-PERF-006	The demo reset shall complete within 30 s (observed: about 6 s).	Should	T	Baseline

4.2 Capacity and Scalability

Table 48: *Capacity requirements*

ID	Requirement	Priority	Verify	Status
NFR-CAP-001	The platform shall support 150 concurrently reporting vehicles at design peak load on the reference deployment.	Must	T	Baseline
NFR-CAP-002	Kafka shall retain 31 days of all platform topics, deleted by age only; disk shall be sized for 31 days at design peak load plus 30 % head-room.	Must	A	Baseline
NFR-CAP-003	Throughput shall scale by adding partitions and service instances; one consumer per partition per group (3 by default) is the parallelism limit and shall be documented in the deployment guide.	Should	A	Baseline
NFR-CAP-004	Reports shall read at most one summary row per vehicle per day (about 4,650 rows for a 150-vehicle month), however many positions exist.	Must	I	Baseline
NFR-CAP-005	The platform should support 1,500 vehicles at the same per-vehicle rate on a 3-broker cluster with 12 partitions per topic.	Could	T	Future

4.3 Reliability and Data Integrity

Table 49: *Reliability requirements*

ID	Requirement	Priority	Verify	Status
NFR-REL-001	No frame acknowledged by the ingest journal shall be lost because of a Kafka outage, service crash, kill -9 or power cut (RPO = 0 for acknowledged frames), provided the host's disk survives.	Must	T	Baseline
NFR-REL-002	Producers shall use acks=all and idempotence; consumers shall commit offsets only after their work is done (at-least-once) and read with isolation read_committed.	Must	I	Baseline
NFR-REL-003	After a service restart, processing shall resume from the committed offsets within 2 minutes, without loss and without duplicate rows.	Must	T	Proposed
NFR-REL-004	With docker-compose.ha.yml (3 brokers, replication factor 3, min.insync.replicas 2), the loss of one broker shall not stop ingestion or processing and shall lose no acknowledged record.	Should	T	Baseline
NFR-REL-005	The database shall be recoverable to any point within the last 31 days by restoring a backup and re-consuming the Kafka topics from that point (see DR-008).	Should	D	Proposed

4.4 Availability

Table 50: *Availability requirements*

ID	Requirement	Priority	Verify	Status
NFR-AVL-001	In a production deployment, the console and the ingestion path should each be available 99.5 % of each calendar month, excluding announced maintenance of at most 4 hours per month, measured by an external probe of /actuator/health every minute.	Should	T	Proposed
NFR-AVL-002	All containers shall restart automatically (`restart: always`) and the service shall start only after PostgreSQL, Redis and Kafka report healthy.	Must	I	Baseline
NFR-AVL-003	/actuator/health shall expose components db, redis, kafkaStreams and dataSafety.	Must	T	Baseline

4.5 Security

Table 51: *Security requirements*

ID	Requirement	Priority	Verify	Status
NFR-SEC-001	No password, key or token shall be committed to the repository; .gitignore shall exclude local-secrets.properties and .env.	Must	I	Baseline
NFR-SEC-002	The service container shall run as a non-root user (uid 1001) from a JRE-only image.	Must	I	Baseline
NFR-SEC-003	The Actuator shall expose only health and info over HTTP.	Must	I	Baseline
NFR-SEC-004	Session cookies shall be HttpOnly, SameSite=Lax, and Secure when served over HTTPS.	Must	T	Baseline
NFR-SEC-005	The smart_truck connection shall use a read-only user and a read-only pool.	Must	I	Baseline
NFR-SEC-006	Outside local development, the console shall refuse to start when DASHBOARD_PASSWORD or AUTH_SECRET is unset or equals the development default.	Must	T	Proposed

ID	Requirement	Priority	Verify	Status
NFR-SEC-007	The service API (port 8080) shall not be reachable from outside the host or private network, or shall require authentication (for example a service token issued to the console).	Must	T	Proposed
NFR-SEC-008	All traffic between browser and console, and across hosts (HP home server), shall use TLS 1.2 or later in production.	Must	T	Proposed
NFR-SEC-009	Every state-changing operation (rules, assumptions, connections, replay, discard, clear, reset, simulator) should be written to an audit log with user, time, old and new value, kept for at least 180 days.	Should	T	Proposed
NFR-SEC-010	Demo reset shall be disabled (`gps.admin.reset-enabled=false`) in every non-demo deployment.	Must	I	Proposed
NFR-SEC-011	Dependencies should be scanned for known vulnerabilities before each release; no release shall ship with a known critical vulnerability without a recorded exception.	Should	I	Proposed

4.6 Privacy

Table 52: *Privacy requirements*

ID	Requirement	Priority	Verify	Status
NFR-PRV-001	Location data linked to an identifiable driver shall be treated as personal data (see REG-001); access shall be limited to signed-in users.	Must	I	Baseline
NFR-PRV-002	The repository, documentation and examples shall contain only dummy IMEIs, registration numbers and company names.	Must	I	Baseline
NFR-PRV-003	Retention periods for positions, events and archives shall be defined per tenant and enforced by a scheduled deletion job (OI-04).	Should	T	Proposed
NFR-PRV-004	Personal data shall not be placed in URL query strings that are logged, other than device and vehicle identifiers.	Should	I	Proposed

4.7 Usability and Accessibility

Table 53: *Usability requirements*

ID	Requirement	Priority	Verify	Status
NFR-USE-001	Every computed figure in Reports shall have an explanation of its formula with its own values (FR-RPT-007).	Must	D	Baseline
NFR-USE-002	Every rejection, dead letter and failure shall be shown with a plain-English reason that names the field and value at fault.	Must	I	Baseline
NFR-USE-003	A new operator should be able to find a named vehicle's live position, its trips for yesterday and its open alerts within 3 minutes without help, in a usability test with at least 5 users.	Should	T	Proposed
NFR-USE-004	The console should meet WCAG 2.1 level AA for colour contrast, keyboard operation and text alternatives in both light and dark appearance.	Should	T	Proposed

4.8 Maintainability and Observability

Table 54: Maintainability requirements

ID	Requirement	Priority	Verify	Status
NFR-MNT-001	Every processing rule in gps-processing and every topology in gps-streams shall have unit tests that run without Kafka, a database or Spring (`mvnw test`).	Must	I	Baseline
NFR-MNT-002	Each Maven module shall have one responsibility (Table 10); Kafka code and business logic shall meet only through handler interfaces.	Must	I	Baseline
NFR-MNT-003	Every consumer stage shall publish counters (total, per second, failures, per IMEI), mirrored to Redis every 5 s.	Must	T	Baseline
NFR-MNT-004	Line coverage of gps-processing and gps-streams should be at least 80 %, measured by JaCoCo in the build.	Should	T	Proposed
NFR-MNT-005	Service logs should be shipped to a central store and kept for 180 days (see REG-003).	Should	I	Proposed

4.9 Compatibility and Portability

Table 55: Compatibility requirements

ID	Requirement	Priority	Verify	Status
NFR-CMP-001	The platform shall run with `docker compose up -d --build` on any Docker host, and natively on Windows 11 with the scripts in infrastructure/windows.	Must	D	Baseline
NFR-CMP-002	The console shall work in the two most recent major versions of Chrome, Edge and Firefox at 1280 × 720 pixels or larger.	Must	T	Proposed
NFR-CMP-003	All REST endpoints shall exchange JSON in UTF-8 with ISO-8601 UTC timestamps.	Must	I	Baseline

4.10 Regulatory and Compliance Requirements

The following regulations apply to a telematics platform that processes vehicle location in India. Applicability and the exact obligations shall be confirmed by legal counsel before production use (OI-05). These requirements do not replace legal advice.

Table 56: Regulatory requirements

ID	Requirement	Priority	Verify	Status
REG-001	Digital Personal Data Protection Act, 2023 [R12]: where positions can identify a driver, the operating tenant (data fiduciary) shall give notice of the purpose, obtain a lawful basis, allow access and erasure requests, and erase data when the purpose is served; the platform shall provide per-tenant export and deletion of a vehicle's or driver's data.	Must	I	Proposed
REG-002	IT Act 2000, section 43A and SPDI Rules 2011 [R13]: the operator shall implement and document reasonable security practices (for example an ISO/IEC 27001-aligned control set) covering NFR-SEC-001 to NFR-SEC-011.	Must	I	Proposed
REG-003	CERT-In Directions of 28 April 2022 [R14]: logs of the platform's ICT systems shall be kept securely for a rolling 180 days within Indian jurisdiction, and cyber-security incidents shall be reportable to CERT-In within 6 hours of being noticed.	Must	I	Proposed

ID	Requirement	Priority	Verify	Status
REG-004	AIS-140 [R15]: when devices fitted under the AIS-140 mandate are connected, the device parser shall accept their data format; the platform is not an emergency-response or state-backend system and shall not be presented as one.	Should	I	Future
REG-005	Third-party terms: before commercial use, the operator shall hold a licence that permits it for the Open-Meteo API and the TomTom API, shall attribute OpenStreetMap data (ODbL) wherever Valhalla routes or map data are shown, and shall review the Redis licence (Redis 7.4 and later are not BSD-licensed) for hosted use.	Must	I	Proposed
REG-006	Speed limits: default and per-tenant speed rules shall be configurable to the limits that apply to the fleet's vehicle class and roads; the default of 60 km/h is a company policy value, not a statutory limit (BR-010).	Must	I	Baseline

5 External Interface Requirements

5.1 User Interfaces

Table 57: Console sections

Section	Main pages
Live	Fleet overview, live map, vehicle list, vehicle inspector, throughput, event stream, fleet simulator.
Processing	Pipeline overview, consumer groups, Kafka Streams (topology and tasks, vehicle table, activity windows, shadow parity), speed violation (live episodes, rules, distance cross-check, events), vehicle state, stoppage and ignition, geofencing*, waypoints*, cumulative distance.
Alerts	All alerts, escalations*, over-speed (dual verification, device log audit, alert policy), trip detection (trips, master table lookup, stops), route deviation*.
Storage	Ingestion overview, schema validation, rejected frames, PostgreSQL, ClickHouse*, workers and back-pressure, data safety, dead letters.
Analytics	Live telemetry, latency, violations and anomalies, trip efficiency.
Notifications*	Inbox, channels, routing rules, delivery log, quiet hours and digest.
Developer	Engine overview, API reference, connections, ingestion, modules, processors, rules, settings, demo tools.
Reports	Day report, month report, vehicle day sheet, fuel and savings, cost assumptions, clear database.

* Planned module: pages show example data, labelled as such (FR-UI-002).

Table 58: User interface requirements

ID	Requirement	Priority	Verify	Status
IFR-UI-001	The console shall be served on port 3000 and forward /api/* to the service at GPS_API_URL (default http://localhost:8080), adding no-store caching headers.	Must	T	Baseline
IFR-UI-002	Destructive actions (clear database, demo reset, discard, connection switch) shall state what will be lost or kept and ask for an explicit confirmation before they run.	Must	D	Proposed

5.2 Software Interfaces: REST API

Base URL: http://<service-host>:8080. Content type application/json; UTF-8. In the baseline, errors are returned as JSON {"error": message} with HTTP 400 (invalid report or connection parameter), 409 (operation not possible in the current state, for example discarding a dead letter that is not parked, or a refused Redis switch) or 503 (Kafka Streams state not yet available); the console proxy answers 502 when the service cannot be reached. Other failures fall back to the framework's default error response; uniform handling is required by IFR-API-001. The API is unversioned; a version prefix is proposed in OI-06.

Table 59: REST endpoints

Method	Path	Purpose
GET	/api/engine/overview	Producers, topics, consumers, lag (dashboard)
GET	/api/engine/devices?imei=	Devices filtered by IMEI fragment
GET	/api/engine/devices/{imei}	One device: partition, per-stage counts, live position
POST	/api/simulator/start?vehicleCount&pingsPerVehicle	Start the fleet
POST	/api/simulator/stop	Stop the fleet
GET	/api/simulator/status	Simulator state and failure reason

Method	Path	Purpose
POST	/api/simulator/send-single-frame	Send one raw frame (body: raw frame JSON)
GET	/api/gps-positions/count	Number of stored positions
GET	/api/gps-positions/recently-stored	Newest 20 stored positions
GET	/api/vehicles/{registrationNo}/latest-position	Latest stored position of a vehicle
GET	/api/trips?imei&from&to	Trips overlapping the period (default last 24 h)
GET	/api/stops?imei&from&to	Stops overlapping the period
GET	/api/trips/count	Open and closed trips and stops
GET	/api/events?type&imei&status&limit	Vehicle events
GET	/api/events/summary	Events per type: last 24 h and open
GET / PUT / DELETE	/api/speed/rules, /api/speed/rules/{scope}	Speed rules (default, tenant:{code}, vehicle:{reg})
GET	/api/speed/active	Vehicles currently above their limit
GET	/api/vehicles/{imei}/state?history=n	Full tracking state of a vehicle
GET	/api/vehicles/states	Vehicle-state hash of every vehicle
GET	/api/streams/overview | vehicles | vehicles/{imei} | fleet/activity | parity	Kafka Streams state (Interactive Queries)
GET	/api/data-safety	Data-safety report
GET	/api/dead-letters?status&topic, /summary, /{id}	Dead letters
POST	/api/dead-letters/{id}/replay, /replay?topic=	Replay one or the oldest parked
POST	/api/dead-letters/{id}/discard?note=	Discard with a note
GET	/api/rejected-frames?imei, /summary	Rejected-frame archive
GET / PUT / POST	/api/connections, /redis, /redis/test, /valhalla, /valhalla/test, /ingestion-api/test	External connections
GET	/api/reports/filters | day | month | vehicle-day	Reports
POST	/api/reports/refresh	Rebuild summaries
GET / PUT	/api/reports/assumptions	Cost assumptions
POST	/api/admin/reset?keepReports&restartFleet	Clear database / demo reset
GET	/actuator/health	Health: db, redis, kafkaStreams, dataSafety

Table 60: *API requirements*

ID	Requirement	Priority	Verify	Status
IFR-API-001	Every endpoint shall validate its parameters and answer an invalid one with HTTP 400 and a JSON message naming the parameter; unexpected failures shall return HTTP 500 with a correlation id and no stack trace.	Must	T	Proposed
IFR-API-002	Time parameters (from, to) shall be ISO-8601 instants; date and month parameters shall be ISO-8601 local dates in the report zone.	Must	T	Baseline
IFR-API-003	List endpoints shall limit their result size (for example events: at most 500) to protect the service.	Must	T	Baseline
IFR-API-004	Discard shall be safe to retry: repeating it on a row that is no longer PARKED shall change nothing and answer HTTP 409. Replay of a REPLAYED row sends the record again and increases replay_count, so a client shall read the row's status before retrying a replay whose outcome is unknown.	Must	T	Baseline
IFR-API-005	A machine-readable API description (OpenAPI 3) should be published with each release.	Should	I	Proposed

5.3 Software Interfaces: Kafka Topics and Messages

Table 61: *Kafka topics*

Topic	Value	Producers	Consumer groups
gps.raw-device-frames	Raw frame JSON	Simulator / parser via journal	gps-raw-frame-processing-service
gps.processed-positions	Processed position JSON (EOS)	Normaliser	storage, live-location, trip, speed, state, stoppage, ignition, gps-streams-analytics
gps.rejected-frames	Rejected frame JSON (EOS)	Normaliser	gps-rejected-frame-logger
gps.vehicle-events	Vehicle event JSON	Stream consumers via journal	(downstream applications)
gps.vehicle-activity	Activity window JSON	gps-streams-analytics	(downstream applications)
gps.dead-letters	Original bytes + error headers	All consumers and Streams apps	gps-dead-letter-archive
gps.shadow.processed-positions, gps.shadow.rejected-frames	As above	Shadow normaliser	Parity join

All topics: key = IMEI (UTF-8 string), 3 partitions, retention 31 days by time only, compression zstd. Changelog topics of the Kafka Streams stores are compacted.

Table 62: *Raw frame (topic gps.raw-device-frames)*

Field	Type	Required	Rule
imei	string	Yes	Non-blank; equals the record key
recordedAt	ISO-8601 instant	Yes	Not more than 5 minutes in the future
latitude	number	Yes	−90 to 90; not 0 together with longitude 0
longitude	number	Yes	−180 to 180
speedKmph	integer	Yes	0 to 150
ignition	boolean	No	Device ignition input when wired
odometerMeters	integer	No	Device odometer; used for distance when believable

Table 63: *Processed position (topic gps.processed-positions)*

Field	Type	Meaning
imei, recordedAt, latitude, longitude, speedKmph, ignition, odometerMeters	as raw frame	Copied from the frame (position may be jitter-clamped)
movementStatus	MOVING | STOPPED	FR-NRM-009
distanceFromPreviousMeters	number ≥ 0	Counted distance of this reading
secondsSincePreviousReading	integer ≥ 0	0 for first and out-of-order readings
arrivedOutOfOrder	boolean	FR-NRM-003
totalDistanceMeters	number	Running total per device
distanceSource	enum	FIRST_READING, GPS, DEVICE_ODOMETER, JITTER_FILTERED, JUMP_FILTERED, OUT_OF_ORDER
impliedSpeedKmph	number	Distance ÷ time to the previous reading
processedAt	ISO-8601 instant	When the normaliser produced it

Table 64: *Rejected frame (topic gps.rejected-frames)*

Field	Type	Meaning
imei	string or null	IMEI if it could be read
originalPayload	string	The frame exactly as received
rejectionReason	string	Reason text from Section 3.4 or 3.5
rejectedAt	ISO-8601 instant	Time of rejection

Table 65: *Vehicle event types*

Type	Raised by	Meaning
OVERSPEED	gps-speed-violation-service	Sustained speed above the applicable limit
SPEED_SENSOR_MISMATCH	gps-speed-violation-service	Speedometer disagrees with GPS speed repeatedly
STOPPAGE	gps-stoppage-service	Stop with engine on; cause verdict attached
PARKED	gps-stoppage-service	Stop with engine off
IDLING	gps-ignition-service	Engine on while standing beyond the limit
IGNITION_ON / IGNITION_OFF	gps-ignition-service	Engine state change (device or virtual)
HARSH_ACCELERATION / HARSH_BRAKING	gps-vehicle-state-service	Acceleration beyond threshold between close readings

Table 66: *Kafka interface requirements*

ID	Requirement	Priority	Verify	Status
IFR-KAF-001	Any producer of raw frames (including a future device parser) shall publish the JSON of Table 62 to gps.raw-device-frames with key = IMEI, through the ingest journal.	Must	T	Baseline
IFR-KAF-002	Topics shall be declared by the service at start-up; automatic topic creation on the broker shall be disabled.	Must	I	Baseline
IFR-KAF-003	Message field names and types shall be changed only in a backward-compatible way (new optional fields); a breaking change requires a new topic name.	Must	I	Proposed

5.4 Software Interfaces: Data Stores and Third-Party Services

Table 67: *Data stores and services*

Interface	Protocol	Use	Failure handling
PostgreSQL gps_tracking	JDBC	System of record (see [R4])	Retry without limit
Redis	RESP	Live positions, state, history, rules, engine counters	Retry; switchable at run time
smart_truck MySQL	JDBC, read-only	SELECT latest trips and pings	Simulator status shows the error
Valhalla	HTTP JSON	Route computation and test	5 s timeout
HP ingestion API	HTTP JSON	Health only (not part of the pipeline)	5 s timeout
Open-Meteo	HTTPS JSON	Weather at a stoppage	3 s timeout; verdict without it
TomTom (optional)	HTTPS JSON	Traffic at a stoppage (key TOMTOM_API_KEY)	3 s timeout; verdict without it

Table 68: Redis key space

Key	Type	Content
gps:live:vehicle:{imei}	hash	Newest position of the vehicle
gps:live:positions	geo set	One point per device, for radius queries
gps:state:vehicle:{imei}	hash	Vehicle state (heading, acceleration, Kalman track)
gps:state:speed:{imei} / ignition / stoppage	hash	Per-consumer working state
gps:history:{imei}	stream	Recent history, about 1000 entries
gps:events:{imei}	list	Recent events of the vehicle
gps:context:weather:{cell}	string	Cached weather of a 0.1° cell (10 min)
gps:config:speed-rule:{scope}, gps:config:speed-rules	hash / set	Speed rules and their scopes (kept by resets)
gps:engine:stage:{stage}[:by-imei]	hash	Stage counters mirrored every 5 s

Table 69: Store interface requirements

ID	Requirement	Priority	Verify	Status
IFR-RDS-001	The platform shall write only keys under the prefixes listed in Table 68 and shall never delete keys outside them.	Must	T	Baseline
IFR-RDS-002	On the HP home server the platform shall use Redis database 1, leaving database 0 to the ingestion API.	Must	I	Baseline

5.5 Hardware and Communication Interfaces

- No hardware is accessed directly. GPS devices are reached through a future parser (IFR-KAF-001). The HP home server is reached over its Wi-Fi hotspot at 192.168.137.1 (Valhalla :8002, Redis :6379, FastAPI :8000).
- Ports in the Docker deployment: console 3000, service 8080, PostgreSQL 5433, Kafka 29092, Redis 6380, RedisInsight 8001 (host side). Containers address each other by service name.
- All timestamps on the wire are UTC ISO-8601; all text is UTF-8.

6 Data and Information Requirements

The logical and physical data model is specified in the companion Database Schema Specification [R4]: entities, columns, types, keys, constraints, indexes, retention and the full DDL. This section states the requirements that the model satisfies.

6.1 Data Entities

Table 70: *Data entities*

Entity (table)	Owner (writer)	Purpose
tenant, vehicle, device, device_assignment	Master data (simulator seeder; future admin API)	Who owns what, and which device was in which vehicle when
gps_position	gps-position-storage-service	Every accepted reading
vehicle_motion_state, gps_trip, gps_stop, gps_trip_late_reading	gps-trip-service	Trip detection state and results
vehicle_event	Stream consumers	Driving and engine events
vehicle_day_summary, report_setting	Report job; Reports UI	Reports and cost assumptions
dead_letter, rejected_frame	Archive consumers	Permanent record of what did not become a position
connection_setting	Connections UI	Run-time external addresses
app_user, auth_key	Portal administration; migration	Who may sign in, as which role; the token signing key
place, shipment, shipment_import	Geofence editor and import; shipment upload; module shipment-tracking	Geofence master, trips of trucks, upload history
geofence_visit	Module geofence	Visits and drive-throughs of geofences
route_plan, route_stop	RoutePlanService; modules route-deviation and waypoints	Planned road routes, their live numbers and stops

6.2 Data Requirements

Table 71: *Data requirements*

ID	Requirement	Priority	Verify	Status
DR-001	Each entity shall have a primary key; uniqueness rules BR-001, BR-002, BR-003 and BR-005 shall be enforced by database constraints, not only by application code.	Must	I	Baseline
DR-002	Telemetry tables shall refer to master data by IMEI, tenant code and registration number (not foreign keys), so that telemetry can be written at high rate and kept when master data changes.	Must	I	Baseline
DR-003	Coordinates shall be stored in WGS-84 decimal degrees (double precision) with range checks; speeds in km/h; distances in metres.	Must	I	Baseline
DR-004	Every timestamp shall be stored as timestamptz in UTC; the JDBC session time zone shall be UTC.	Must	I	Baseline
DR-005	Enumerated values (status, kind, movement status) shall be checked by database CHECK constraints.	Must	I	Baseline
DR-006	A trip's distance shall satisfy distance_m ≥ 0 and end ≥ start (database CHECK constraints).	Must	T	Baseline

ID	Requirement	Priority	Verify	Status
DR-007	Kafka shall keep all platform topics for 31 days; PostgreSQL rows shall be kept until deleted by a defined retention policy (NFR-PRV-003); archives shall be kept indefinitely.	Must	I	Baseline
DR-008	PostgreSQL should be backed up at least daily, with backups kept for 31 days and a restore tested at least each quarter; recovery may re-consume Kafka from the backup time (NFR-REL-005).	Should	D	Proposed
DR-009	Concurrent updates of the same trip state shall be prevented by the guarded update (FR-TRP-008); concurrent edits of configuration shall be last-writer-wins and audited (NFR-SEC-009).	Must	T	Baseline
DR-010	Report summaries frozen by Clear database (generation > 0) shall never be recomputed or altered.	Must	T	Baseline

7 Verification and Acceptance

7.1 Verification Approach

- **Unit tests** (T): every rule in gps-processing and gps-streams, run by `mvnw test` without infrastructure. Existing test classes are listed in the RTM.
- **Integration tests** (T): the Docker deployment with the simulator, checking database rows, Redis keys and API answers.
- **Failure-injection tests** (T): stop Kafka, PostgreSQL or Redis, or kill the service, under load; then compare counts (sent = stored + rejected + dead-lettered).
- **Performance tests** (T): design peak load for 15 minutes; collect consumer lag every second and timestamps from gps_position.
- **Inspections** (I): code, configuration and document reviews against the requirement text.
- **Demonstrations** (D): console walkthroughs observed by the product owner.

7.2 Acceptance Criteria

Table 72: *Acceptance criteria*

ID	Scenario and expected result	Covers
AC-01	Start 150 vehicles. Within 10 minutes all consumer groups are STABLE, lag returns to 0 after catch-up, failures are 0, and Redis holds 150 live positions.	NFR-CAP-001, NFR-PERF-001, FR-LIVE-001
AC-02	Send a frame with latitude 95.0. It is rejected with reason “Latitude 95.0 is outside -90..90”, appears in rejected_frame and in the log with its IMEI.	FR-VAL-006, FR-DLQ-007
AC-03	Stop Kafka for 5 minutes while the fleet runs, then start it. The number of stored positions equals the number of accepted frames sent; no gap in any device's sequence.	FR-ING-004, NFR-REL-001
AC-04	Kill the service (kill -9) under load and restart it. No acknowledged frame is lost and no (imei, recorded_at) is stored twice.	FR-ING-005, FR-STO-002, NFR-REL-003
AC-05	Replay the reference drive (test resource osmand-sent-positions.jsonl). The trip starts at reading 10 and stops at reading 40, as in the unit test.	FR-TRP-001, FR-TRP-002
AC-06	Deliver a 5-minute-late reading during a trip. Trip distance does not change and late_reading_count increases by 1.	FR-TRP-007, BR-011
AC-07	Set a vehicle speed rule of 40 km/h. A vehicle driving above 40 km/h for 60 s raises exactly one OVERSPEED event, CLOSED after the speed drops below 35 km/h.	FR-SPD-001 to 003
AC-08	Corrupt a record so that the storage consumer cannot parse it. It is parked within 1 s, archived as PARKED, later records are stored, and a replay after the fix stores it.	FR-DLQ-001, 004, 005
AC-09	Open the day report for yesterday. Every figure's explanation reproduces the figure from the stated formula and cost assumptions; change the fuel price and the cost changes at once.	FR-RPT-004, 007, 008
AC-10	Clear the database with “keep reports”. Positions are gone; yesterday's day report shows the same totals as before.	FR-RPT-010, DR-010
AC-11	Try 5 wrong passwords. The sixth attempt within 60 s is refused; an /api call without a session returns 401.	FR-AUTH-001, FR-AUTH-005
AC-12	Switch normalisation from owner to shadow and back under load. Parity shows MATCH for at least 99.9 % of readings; no reading is skipped or processed twice.	FR-KS-001, FR-KS-002, CON-007

The release is accepted when all “Must” requirements with status Baseline pass, every acceptance criterion passes, and every failed “Should” has a recorded decision.

8 Traceability

8.1 Traceability Approach

Each requirement traces back to a business objective (Table 4) and forward to the design component that implements it and the test that verifies it. Test case TC-<requirement ID> is the planned test for each requirement; where an automated test already exists it is named. The matrix is generated from the requirement tables of this document, so it cannot drift from them.

8.2 Requirements Traceability Matrix

Table 73: *Requirements traceability matrix*

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
CON-001	OBJ-07	Architecture	TC-CON-001; Code review	Baseline
CON-002	OBJ-07	Architecture	TC-CON-002; Code review	Baseline
CON-003	OBJ-07	Architecture	TC-CON-003; Code review	Baseline
CON-004	OBJ-07	Architecture	TC-CON-004; Code review	Baseline
CON-005	OBJ-07	Architecture	TC-CON-005; Code review	Baseline
CON-006	OBJ-07	Architecture	TC-CON-006; Code review	Baseline
CON-007	OBJ-07	Architecture	TC-CON-007; Code review	Baseline
CON-008	OBJ-07	Architecture	TC-CON-008; Code review	Baseline
FR-AUTH-001	OBJ-08	gps-dashboard-ui: proxy.ts, session.ts, auth-actions.ts	TC-FR-AUTH-001; Manual (no automated UI tests yet)	Baseline
FR-AUTH-002	OBJ-08	gps-dashboard-ui: proxy.ts, session.ts, auth-actions.ts	TC-FR-AUTH-002; Manual (no automated UI tests yet)	Baseline
FR-AUTH-003	OBJ-08	gps-dashboard-ui: proxy.ts, session.ts, auth-actions.ts	TC-FR-AUTH-003; Manual (no automated UI tests yet)	Baseline
FR-AUTH-004	OBJ-08	gps-dashboard-ui: proxy.ts, session.ts, auth-actions.ts	TC-FR-AUTH-004; Manual (no automated UI tests yet)	Baseline
FR-AUTH-005	OBJ-08	gps-dashboard-ui: proxy.ts, session.ts, auth-actions.ts	TC-FR-AUTH-005; Manual (no automated UI tests yet)	Baseline
FR-AUTH-006	OBJ-08	gps-dashboard-ui: proxy.ts, session.ts, auth-actions.ts	TC-FR-AUTH-006; Manual (no automated UI tests yet)	Baseline
FR-AUTH-007	OBJ-08	gps-dashboard-ui: proxy.ts, session.ts, auth-actions.ts	TC-FR-AUTH-007; Manual (no automated UI tests yet)	Baseline
FR-AUTH-008	OBJ-08	gps-dashboard-ui: proxy.ts, session.ts, auth-actions.ts	TC-FR-AUTH-008; Manual (no automated UI tests yet)	Baseline
FR-AUTH-009	OBJ-08	gps-dashboard-ui: proxy.ts, session.ts, auth-actions.ts	TC-FR-AUTH-009; Manual (no automated UI tests yet)	Baseline
FR-AUTH-010	OBJ-08	gps-dashboard-ui: proxy.ts, session.ts, auth-actions.ts	TC-FR-AUTH-010; Manual (no automated UI tests yet)	Proposed
FR-MD-001	OBJ-01	masterdata/, entity/, V1 migration	TC-FR-MD-001; TenantCodeFormatterTest	Baseline
FR-MD-002	OBJ-01	masterdata/, entity/, V1 migration	TC-FR-MD-002; TenantCodeFormatterTest	Baseline
FR-MD-003	OBJ-01	masterdata/, entity/, V1 migration	TC-FR-MD-003; TenantCodeFormatterTest	Baseline
FR-MD-004	OBJ-01	masterdata/, entity/, V1 migration	TC-FR-MD-004; TenantCodeFormatterTest	Baseline
FR-MD-005	OBJ-01	masterdata/, entity/, V1 migration	TC-FR-MD-005; TenantCodeFormatterTest	Baseline
FR-MD-006	OBJ-01	masterdata/, entity/, V1 migration	TC-FR-MD-006; TenantCodeFormatterTest	Baseline
FR-MD-007	OBJ-01	masterdata/, entity/, V1 migration	TC-FR-MD-007; TenantCodeFormatterTest	Proposed
FR-ING-001	OBJ-02	flow/, journal/, kafka/RawGpsFramePublisher	TC-FR-ING-001; IngestJournalTest, DurableKafkaSenderTest	Baseline
FR-ING-002	OBJ-02	flow/, journal/, kafka/RawGpsFramePublisher	TC-FR-ING-002; IngestJournalTest, DurableKafkaSenderTest	Baseline

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
FR-ING-003	OBJ-02	flow/, journal/, kafka/RawGpsFramePublisher	TC-FR-ING-003; IngestJournalTest, DurableKafkaSenderTest	Baseline
FR-ING-004	OBJ-02	flow/, journal/, kafka/RawGpsFramePublisher	TC-FR-ING-004; IngestJournalTest, DurableKafkaSenderTest	Baseline
FR-ING-005	OBJ-02	flow/, journal/, kafka/RawGpsFramePublisher	TC-FR-ING-005; IngestJournalTest, DurableKafkaSenderTest	Baseline
FR-ING-006	OBJ-02	flow/, journal/, kafka/RawGpsFramePublisher	TC-FR-ING-006; IngestJournalTest, DurableKafkaSenderTest	Baseline
FR-ING-007	OBJ-02	flow/, journal/, kafka/RawGpsFramePublisher	TC-FR-ING-007; IngestJournalTest, DurableKafkaSenderTest	Baseline
FR-ING-008	OBJ-02	flow/, journal/, kafka/RawGpsFramePublisher	TC-FR-ING-008; IngestJournalTest, DurableKafkaSenderTest	Future
FR-VAL-001	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-001; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-VAL-002	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-002; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-VAL-003	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-003; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-VAL-004	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-004; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-VAL-005	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-005; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-VAL-006	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-006; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-VAL-007	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-007; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-VAL-008	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-008; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-VAL-009	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-009; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-VAL-010	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-010; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-VAL-011	OBJ-02	gps-processing rule/GpsFrameValidationRules	TC-FR-VAL-011; GpsFrameValidationRulesTest, NormalizationTopologyTest	Baseline
FR-NRM-001	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-001; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-NRM-002	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-002; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-NRM-003	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-003; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
FR-NRM-004	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-004; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-NRM-005	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-005; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-NRM-006	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-006; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-NRM-007	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-007; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-NRM-008	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-008; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-NRM-009	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-009; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-NRM-010	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-010; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-NRM-011	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-011; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-NRM-012	OBJ-02	gps-processing step/, rule/; gps-streams normalize/	TC-FR-NRM-012; CalculateMovementStepTest, MovementRulesTest, GpsFrameProcessorTest, NormalizationTopologyTest	Baseline
FR-STO-001	OBJ-02	storage/GpsPositionStorageService	TC-FR-STO-001; Integration (TC)	Baseline
FR-STO-002	OBJ-02	storage/GpsPositionStorageService	TC-FR-STO-002; Integration (TC)	Baseline
FR-STO-003	OBJ-02	storage/GpsPositionStorageService	TC-FR-STO-003; Integration (TC)	Baseline
FR-LIVE-001	OBJ-01	livelocation/, redis/LiveVehiclePositionStore, console Live	TC-FR-LIVE-001; Integration (TC)	Baseline
FR-LIVE-002	OBJ-01	livelocation/, redis/LiveVehiclePositionStore, console Live	TC-FR-LIVE-002; Integration (TC)	Baseline
FR-LIVE-003	OBJ-01	livelocation/, redis/LiveVehiclePositionStore, console Live	TC-FR-LIVE-003; Integration (TC)	Baseline
FR-TRP-001	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-001; TripMotionProcessorTest, MotionStateJsonTest	Baseline
FR-TRP-002	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-002; TripMotionProcessorTest, MotionStateJsonTest	Baseline
FR-TRP-003	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-003; TripMotionProcessorTest, MotionStateJsonTest	Baseline
FR-TRP-004	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-004; TripMotionProcessorTest, MotionStateJsonTest	Baseline

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
FR-TRP-005	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-005; TripMotionProcessorTest, MotionStateJsonTest	Baseline
FR-TRP-006	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-006; TripMotionProcessorTest, MotionStateJsonTest	Baseline
FR-TRP-007	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-007; TripMotionProcessorTest, MotionStateJsonTest	Baseline
FR-TRP-008	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-008; TripMotionProcessorTest, MotionStateJsonTest	Baseline
FR-TRP-009	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-009; TripMotionProcessorTest, MotionStateJsonTest	Baseline
FR-TRP-010	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-010; TripMotionProcessorTest, MotionStateJsonTest	Baseline
FR-TRP-011	OBJ-04	gps-processing trip/TripMotionProcessor; trip/TripDetectionService	TC-FR-TRP-011; TripMotionProcessorTest, MotionStateJsonTest	Future
FR-SPD-001	OBJ-03	gps-processing speed/; streaming/SpeedViolationService, SpeedRuleStore	TC-FR-SPD-001; SpeedViolationProcessorTest	Baseline
FR-SPD-002	OBJ-03	gps-processing speed/; streaming/SpeedViolationService, SpeedRuleStore	TC-FR-SPD-002; SpeedViolationProcessorTest	Baseline
FR-SPD-003	OBJ-03	gps-processing speed/; streaming/SpeedViolationService, SpeedRuleStore	TC-FR-SPD-003; SpeedViolationProcessorTest	Baseline
FR-SPD-004	OBJ-03	gps-processing speed/; streaming/SpeedViolationService, SpeedRuleStore	TC-FR-SPD-004; SpeedViolationProcessorTest	Baseline
FR-SPD-005	OBJ-03	gps-processing speed/; streaming/SpeedViolationService, SpeedRuleStore	TC-FR-SPD-005; SpeedViolationProcessorTest	Baseline
FR-SPD-006	OBJ-03	gps-processing speed/; streaming/SpeedViolationService, SpeedRuleStore	TC-FR-SPD-006; SpeedViolationProcessorTest	Baseline
FR-VST-001	OBJ-03	gps-processing state/; streaming/VehicleStateService	TC-FR-VST-001; VehicleStateProcessorTest, StateCodecsTest	Baseline
FR-VST-002	OBJ-03	gps-processing state/; streaming/VehicleStateService	TC-FR-VST-002; VehicleStateProcessorTest, StateCodecsTest	Baseline
FR-VST-003	OBJ-03	gps-processing state/; streaming/VehicleStateService	TC-FR-VST-003; VehicleStateProcessorTest, StateCodecsTest	Baseline
FR-VST-004	OBJ-03	gps-processing state/; streaming/VehicleStateService	TC-FR-VST-004; VehicleStateProcessorTest, StateCodecsTest	Baseline
FR-VST-005	OBJ-03	gps-processing state/; streaming/VehicleStateService	TC-FR-VST-005; VehicleStateProcessorTest, StateCodecsTest	Baseline
FR-STP-001	OBJ-03	gps-processing stoppage/; streaming/StoppageService, StoppageContextAgents	TC-FR-STP-001; StoppageDetectorTest	Baseline
FR-STP-002	OBJ-03	gps-processing stoppage/; streaming/StoppageService, StoppageContextAgents	TC-FR-STP-002; StoppageDetectorTest	Baseline
FR-STP-003	OBJ-03	gps-processing stoppage/; streaming/StoppageService, StoppageContextAgents	TC-FR-STP-003; StoppageDetectorTest	Baseline
FR-STP-004	OBJ-03	gps-processing stoppage/; streaming/StoppageService, StoppageContextAgents	TC-FR-STP-004; StoppageDetectorTest	Baseline
FR-STP-005	OBJ-03	gps-processing stoppage/; streaming/StoppageService, StoppageContextAgents	TC-FR-STP-005; StoppageDetectorTest	Baseline
FR-IGN-001	OBJ-03	gps-processing ignition/; streaming/IgnitionService	TC-FR-IGN-001; IgnitionProcessorTest	Baseline

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
FR-IGN-002	OBJ-03	gps-processing ignition/; streaming/IgnitionService	TC-FR-IGN-002; IgnitionProcessorTest	Baseline
FR-IGN-003	OBJ-03	gps-processing ignition/; streaming/IgnitionService	TC-FR-IGN-003; IgnitionProcessorTest	Baseline
FR-EVT-001	OBJ-03	events/VehicleEventRecorder, kafka/VehicleEventPublisher	TC-FR-EVT-001; Integration (TC)	Baseline
FR-EVT-002	OBJ-03	events/VehicleEventRecorder, kafka/VehicleEventPublisher	TC-FR-EVT-002; Integration (TC)	Baseline
FR-EVT-003	OBJ-03	events/VehicleEventRecorder, kafka/VehicleEventPublisher	TC-FR-EVT-003; Integration (TC)	Baseline
FR-EVT-004	OBJ-03	events/VehicleEventRecorder, kafka/VehicleEventPublisher	TC-FR-EVT-004; Integration (TC)	Baseline
FR-EVT-005	OBJ-03	events/VehicleEventRecorder, kafka/VehicleEventPublisher	TC-FR-EVT-005; Integration (TC)	Future
FR-KS-001	OBJ-02	gps-streams; kafkastreams/	TC-FR-KS-001; NormalizationTopologyTest, AnalyticsTopologyTest, DeadLetterTopologyTest, TopologyShapeTest	Baseline
FR-KS-002	OBJ-02	gps-streams; kafkastreams/	TC-FR-KS-002; NormalizationTopologyTest, AnalyticsTopologyTest, DeadLetterTopologyTest, TopologyShapeTest	Baseline
FR-KS-003	OBJ-02	gps-streams; kafkastreams/	TC-FR-KS-003; NormalizationTopologyTest, AnalyticsTopologyTest, DeadLetterTopologyTest, TopologyShapeTest	Baseline
FR-KS-004	OBJ-02	gps-streams; kafkastreams/	TC-FR-KS-004; NormalizationTopologyTest, AnalyticsTopologyTest, DeadLetterTopologyTest, TopologyShapeTest	Baseline
FR-KS-005	OBJ-02	gps-streams; kafkastreams/	TC-FR-KS-005; NormalizationTopologyTest, AnalyticsTopologyTest, DeadLetterTopologyTest, TopologyShapeTest	Baseline
FR-KS-006	OBJ-02	gps-streams; kafkastreams/	TC-FR-KS-006; NormalizationTopologyTest, AnalyticsTopologyTest, DeadLetterTopologyTest, TopologyShapeTest	Baseline
FR-DLQ-001	OBJ-02	kafka/KafkaRetrySettings, DeadLetter*; archive/	TC-FR-DLQ-001; DeadLetterHandlingTest, DeadLetterTopologyTest	Baseline
FR-DLQ-002	OBJ-02	kafka/KafkaRetrySettings, DeadLetter*; archive/	TC-FR-DLQ-002; DeadLetterHandlingTest, DeadLetterTopologyTest	Baseline
FR-DLQ-003	OBJ-02	kafka/KafkaRetrySettings, DeadLetter*; archive/	TC-FR-DLQ-003; DeadLetterHandlingTest, DeadLetterTopologyTest	Baseline
FR-DLQ-004	OBJ-02	kafka/KafkaRetrySettings, DeadLetter*; archive/	TC-FR-DLQ-004; DeadLetterHandlingTest, DeadLetterTopologyTest	Baseline
FR-DLQ-005	OBJ-02	kafka/KafkaRetrySettings, DeadLetter*; archive/	TC-FR-DLQ-005; DeadLetterHandlingTest, DeadLetterTopologyTest	Baseline
FR-DLQ-006	OBJ-02	kafka/KafkaRetrySettings, DeadLetter*; archive/	TC-FR-DLQ-006; DeadLetterHandlingTest, DeadLetterTopologyTest	Baseline
FR-DLQ-007	OBJ-02	kafka/KafkaRetrySettings, DeadLetter*; archive/	TC-FR-DLQ-007; DeadLetterHandlingTest, DeadLetterTopologyTest	Baseline
FR-DLQ-008	OBJ-02	kafka/KafkaRetrySettings, DeadLetter*; archive/	TC-FR-DLQ-008; DeadLetterHandlingTest, DeadLetterTopologyTest	Baseline
FR-SAFE-001	OBJ-02	safety/DataSafetyChecks, DataSafetyService	TC-FR-SAFE-001; DataSafetyChecksTest	Baseline
FR-SAFE-002	OBJ-02	safety/DataSafetyChecks, DataSafetyService	TC-FR-SAFE-002; DataSafetyChecksTest	Baseline
FR-SAFE-003	OBJ-02	safety/DataSafetyChecks, DataSafetyService	TC-FR-SAFE-003; DataSafetyChecksTest	Baseline
FR-RPT-001	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-001; FuelEstimatorTest	Baseline

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
FR-RPT-002	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-002; FuelEstimatorTest	Baseline
FR-RPT-003	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-003; FuelEstimatorTest	Baseline
FR-RPT-004	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-004; FuelEstimatorTest	Baseline
FR-RPT-005	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-005; FuelEstimatorTest	Baseline
FR-RPT-006	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-006; FuelEstimatorTest	Baseline
FR-RPT-007	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-007; FuelEstimatorTest	Baseline
FR-RPT-008	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-008; FuelEstimatorTest	Baseline
FR-RPT-009	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-009; FuelEstimatorTest	Baseline
FR-RPT-010	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-010; FuelEstimatorTest	Baseline
FR-RPT-011	OBJ-05	report/; gps-processing report/FuelEstimator; console Reports	TC-FR-RPT-011; FuelEstimatorTest	Baseline
FR-CON-001	OBJ-06	connections/, redis/SwitchableRedisConnectionFactory	TC-FR-CON-001; RedisTargetTest, JsonHttpTest	Baseline
FR-CON-002	OBJ-06	connections/, redis/SwitchableRedisConnectionFactory	TC-FR-CON-002; RedisTargetTest, JsonHttpTest	Baseline
FR-CON-003	OBJ-06	connections/, redis/SwitchableRedisConnectionFactory	TC-FR-CON-003; RedisTargetTest, JsonHttpTest	Baseline
FR-CON-004	OBJ-06	connections/, redis/SwitchableRedisConnectionFactory	TC-FR-CON-004; RedisTargetTest, JsonHttpTest	Baseline
FR-CON-005	OBJ-06	connections/, redis/SwitchableRedisConnectionFactory	TC-FR-CON-005; RedisTargetTest, JsonHttpTest	Baseline
FR-CON-006	OBJ-06	connections/, redis/SwitchableRedisConnectionFactory	TC-FR-CON-006; RedisTargetTest, JsonHttpTest	Baseline
FR-SIM-001	OBJ-06	simulator/; smarttruck/SmartTruckGpsReader	TC-FR-SIM-001; Integration (TC)	Baseline
FR-SIM-002	OBJ-06	simulator/; smarttruck/SmartTruckGpsReader	TC-FR-SIM-002; Integration (TC)	Baseline
FR-SIM-003	OBJ-06	simulator/; smarttruck/SmartTruckGpsReader	TC-FR-SIM-003; Integration (TC)	Baseline
FR-SIM-004	OBJ-06	simulator/; smarttruck/SmartTruckGpsReader	TC-FR-SIM-004; Integration (TC)	Baseline
FR-SIM-005	OBJ-06	simulator/; smarttruck/SmartTruckGpsReader	TC-FR-SIM-005; Integration (TC)	Baseline
FR-ADM-001	OBJ-06	engine/, admin/, monitoring/	TC-FR-ADM-001; StageCountersTest	Baseline
FR-ADM-002	OBJ-06	engine/, admin/, monitoring/	TC-FR-ADM-002; StageCountersTest	Baseline
FR-ADM-003	OBJ-06	engine/, admin/, monitoring/	TC-FR-ADM-003; StageCountersTest	Baseline
FR-ADM-004	OBJ-06	engine/, admin/, monitoring/	TC-FR-ADM-004; StageCountersTest	Baseline
FR-ADM-005	OBJ-06	engine/, admin/, monitoring/	TC-FR-ADM-005; StageCountersTest	Baseline
FR-UI-001	OBJ-06	gps-dashboard-ui	TC-FR-UI-001; Manual (TC)	Baseline
FR-UI-002	OBJ-06	gps-dashboard-ui	TC-FR-UI-002; Manual (TC)	Baseline
FR-UI-003	OBJ-06	gps-dashboard-ui	TC-FR-UI-003; Manual (TC)	Baseline
FR-UI-004	OBJ-06	gps-dashboard-ui	TC-FR-UI-004; Manual (TC)	Proposed
FR-TMS-001	OBJ-01	tms/*, auth/*, dashboard portals	TC-FR-TMS-001; PortalSignInTest, ShipmentFileParsingTest	Baseline
FR-TMS-002	OBJ-01	tms/*, auth/*, dashboard portals	TC-FR-TMS-002; PortalSignInTest, ShipmentFileParsingTest	Baseline

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
FR-TMS-003	OBJ-01	tms/*, auth/*, dashboard portals	TC-FR-TMS-003; PortalSignInTest, ShipmentFileParsingTest	Baseline
FR-TMS-004	OBJ-01	tms/*, auth/*, dashboard portals	TC-FR-TMS-004; PortalSignInTest, ShipmentFileParsingTest	Baseline
FR-TMS-005	OBJ-01	tms/*, auth/*, dashboard portals	TC-FR-TMS-005; PortalSignInTest, ShipmentFileParsingTest	Baseline
FR-TMS-006	OBJ-01	tms/*, auth/*, dashboard portals	TC-FR-TMS-006; PortalSignInTest, ShipmentFileParsingTest	Baseline
FR-TMS-007	OBJ-01	tms/*, auth/*, dashboard portals	TC-FR-TMS-007; PortalSignInTest, ShipmentFileParsingTest	Baseline
FR-TMS-008	OBJ-01	tms/*, auth/*, dashboard portals	TC-FR-TMS-008; PortalSignInTest, ShipmentFileParsingTest	Baseline
FR-MOD-001	OBJ-07	gps-module-api; modules/*	TC-FR-MOD-001; Module registry checks in the verify stack	Baseline
FR-MOD-002	OBJ-07	gps-module-api; modules/*	TC-FR-MOD-002; Module registry checks in the verify stack	Baseline
FR-MOD-003	OBJ-07	gps-module-api; modules/*	TC-FR-MOD-003; Module registry checks in the verify stack	Baseline
FR-MOD-004	OBJ-07	gps-module-api; modules/*	TC-FR-MOD-004; Module registry checks in the verify stack	Baseline
FR-MOD-005	OBJ-07	gps-module-api; modules/*	TC-FR-MOD-005; Module registry checks in the verify stack	Baseline
FR-GEO-001	OBJ-03	gps-processing geofence/; geofence/*	TC-FR-GEO-001; GeofenceIndexTest, GeofenceTrackerTest, SmartTruckParityCheck	Baseline
FR-GEO-002	OBJ-03	gps-processing geofence/; geofence/*	TC-FR-GEO-002; GeofenceIndexTest, GeofenceTrackerTest, SmartTruckParityCheck	Baseline
FR-GEO-003	OBJ-03	gps-processing geofence/; geofence/*	TC-FR-GEO-003; GeofenceIndexTest, GeofenceTrackerTest, SmartTruckParityCheck	Baseline
FR-GEO-004	OBJ-03	gps-processing geofence/; geofence/*	TC-FR-GEO-004; GeofenceIndexTest, GeofenceTrackerTest, SmartTruckParityCheck	Baseline
FR-GEO-005	OBJ-03	gps-processing geofence/; geofence/*	TC-FR-GEO-005; GeofenceIndexTest, GeofenceTrackerTest, SmartTruckParityCheck	Baseline
FR-GEO-006	OBJ-03	gps-processing geofence/; geofence/*	TC-FR-GEO-006; GeofenceIndexTest, GeofenceTrackerTest, SmartTruckParityCheck	Baseline
FR-GEO-007	OBJ-03	gps-processing geofence/; geofence/*	TC-FR-GEO-007; GeofenceIndexTest, GeofenceTrackerTest, SmartTruckParityCheck	Baseline
FR-GEO-008	OBJ-03	gps-processing geofence/; geofence/*	TC-FR-GEO-008; GeofenceIndexTest, GeofenceTrackerTest, SmartTruckParityCheck	Baseline
FR-GEO-009	OBJ-03	gps-processing geofence/; geofence/*	TC-FR-GEO-009; GeofenceIndexTest, GeofenceTrackerTest, SmartTruckParityCheck	Baseline
FR-SPD-007	OBJ-03	speed/SpeedPolicy; streaming/SpeedViolationService	TC-FR-SPD-007; SpeedPolicyTest	Baseline
FR-SPD-008	OBJ-03	speed/SpeedPolicy; streaming/SpeedViolationService	TC-FR-SPD-008; SpeedPolicyTest	Baseline
FR-SPD-009	OBJ-03	speed/SpeedPolicy; streaming/SpeedViolationService	TC-FR-SPD-009; SpeedPolicyTest	Baseline
FR-RTE-001	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-001; RouteTrackerTest	Baseline
FR-RTE-002	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-002; RouteTrackerTest	Baseline

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
FR-RTE-003	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-003; RouteTrackerTest	Baseline
FR-RTE-004	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-004; RouteTrackerTest	Baseline
FR-RTE-005	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-005; RouteTrackerTest	Baseline
FR-RTE-006	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-006; RouteTrackerTest	Baseline
FR-RTE-007	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-007; RouteTrackerTest	Baseline
FR-RTE-008	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-008; RouteTrackerTest	Baseline
FR-RTE-009	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-009; RouteTrackerTest	Baseline
FR-RTE-010	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-010; RouteTrackerTest	Baseline
FR-RTE-011	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-011; RouteTrackerTest	Baseline
FR-RTE-012	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-012; RouteTrackerTest	Baseline
FR-RTE-013	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-013; RouteTrackerTest	Baseline
FR-RTE-014	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-014; RouteTrackerTest	Baseline
FR-RTE-015	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-015; RouteTrackerTest	Future
FR-RTE-016	OBJ-03	gps-processing route/; route/*; dashboard components/routes	TC-FR-RTE-016; RouteTrackerTest	Future
FR-WPT-001	OBJ-03	gps-processing route/WaypointTracker; route/WaypointModule	TC-FR-WPT-001; WaypointTrackerTest	Baseline
FR-WPT-002	OBJ-03	gps-processing route/WaypointTracker; route/WaypointModule	TC-FR-WPT-002; WaypointTrackerTest	Baseline
FR-WPT-003	OBJ-03	gps-processing route/WaypointTracker; route/WaypointModule	TC-FR-WPT-003; WaypointTrackerTest	Baseline
FR-WPT-004	OBJ-03	gps-processing route/WaypointTracker; route/WaypointModule	TC-FR-WPT-004; WaypointTrackerTest	Baseline
FR-WPT-005	OBJ-03	gps-processing route/WaypointTracker; route/WaypointModule	TC-FR-WPT-005; WaypointTrackerTest	Baseline
FR-WPT-006	OBJ-03	gps-processing route/WaypointTracker; route/WaypointModule	TC-FR-WPT-006; WaypointTrackerTest	Baseline
BR-001	OBJ-02	Database constraints and processing rules	TC-BR-001; Schema and unit tests	Baseline
BR-002	OBJ-02	Database constraints and processing rules	TC-BR-002; Schema and unit tests	Baseline
BR-003	OBJ-02	Database constraints and processing rules	TC-BR-003; Schema and unit tests	Baseline
BR-004	OBJ-02	Database constraints and processing rules	TC-BR-004; Schema and unit tests	Baseline
BR-005	OBJ-02	Database constraints and processing rules	TC-BR-005; Schema and unit tests	Baseline
BR-006	OBJ-02	Database constraints and processing rules	TC-BR-006; Schema and unit tests	Baseline
BR-007	OBJ-02	Database constraints and processing rules	TC-BR-007; Schema and unit tests	Baseline
BR-008	OBJ-02	Database constraints and processing rules	TC-BR-008; Schema and unit tests	Baseline
BR-009	OBJ-02	Database constraints and processing rules	TC-BR-009; Schema and unit tests	Baseline
BR-010	OBJ-02	Database constraints and processing rules	TC-BR-010; Schema and unit tests	Baseline
BR-011	OBJ-02	Database constraints and processing rules	TC-BR-011; Schema and unit tests	Baseline
BR-012	OBJ-02	Database constraints and processing rules	TC-BR-012; Schema and unit tests	Baseline
BR-013	OBJ-02	Database constraints and processing rules	TC-BR-013; Schema and unit tests	Baseline
BR-014	OBJ-02	Database constraints and processing rules	TC-BR-014; Schema and unit tests	Baseline

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
BR-015	OBJ-02	Database constraints and processing rules	TC-BR-015; Schema and unit tests	Baseline
BR-016	OBJ-02	Database constraints and processing rules	TC-BR-016; Schema and unit tests	Baseline
BR-017	OBJ-02	Database constraints and processing rules	TC-BR-017; Schema and unit tests	Baseline
NFR-PERF-001	OBJ-01	Whole pipeline	TC-NFR-PERF-001; Performance test (TC)	Proposed
NFR-PERF-002	OBJ-01	Whole pipeline	TC-NFR-PERF-002; Performance test (TC)	Proposed
NFR-PERF-003	OBJ-01	Whole pipeline	TC-NFR-PERF-003; Performance test (TC)	Proposed
NFR-PERF-004	OBJ-01	Whole pipeline	TC-NFR-PERF-004; Performance test (TC)	Proposed
NFR-PERF-005	OBJ-01	Whole pipeline	TC-NFR-PERF-005; Performance test (TC)	Proposed
NFR-PERF-006	OBJ-01	Whole pipeline	TC-NFR-PERF-006; Performance test (TC)	Baseline
NFR-CAP-001	OBJ-07	Kafka, report/	TC-NFR-CAP-001; Analysis, performance test	Baseline
NFR-CAP-002	OBJ-07	Kafka, report/	TC-NFR-CAP-002; Analysis, performance test	Baseline
NFR-CAP-003	OBJ-07	Kafka, report/	TC-NFR-CAP-003; Analysis, performance test	Baseline
NFR-CAP-004	OBJ-07	Kafka, report/	TC-NFR-CAP-004; Analysis, performance test	Baseline
NFR-CAP-005	OBJ-07	Kafka, report/	TC-NFR-CAP-005; Analysis, performance test	Future
NFR-REL-001	OBJ-02	journal/, kafka/, docker-compose	TC-NFR-REL-001; DurableKafkaSenderTest, IngestJournalTest, failure-injection test (TC)	Baseline
NFR-REL-002	OBJ-02	journal/, kafka/, docker-compose	TC-NFR-REL-002; DurableKafkaSenderTest, IngestJournalTest, failure-injection test (TC)	Baseline
NFR-REL-003	OBJ-02	journal/, kafka/, docker-compose	TC-NFR-REL-003; DurableKafkaSenderTest, IngestJournalTest, failure-injection test (TC)	Proposed
NFR-REL-004	OBJ-02	journal/, kafka/, docker-compose	TC-NFR-REL-004; DurableKafkaSenderTest, IngestJournalTest, failure-injection test (TC)	Baseline
NFR-REL-005	OBJ-02	journal/, kafka/, docker-compose	TC-NFR-REL-005; DurableKafkaSenderTest, IngestJournalTest, failure-injection test (TC)	Proposed
NFR-AVL-001	OBJ-06	docker-compose.yml, Actuator	TC-NFR-AVL-001; Inspection, health test	Proposed
NFR-AVL-002	OBJ-06	docker-compose.yml, Actuator	TC-NFR-AVL-002; Inspection, health test	Baseline
NFR-AVL-003	OBJ-06	docker-compose.yml, Actuator	TC-NFR-AVL-003; Inspection, health test	Baseline
NFR-SEC-001	OBJ-08	Deployment, console, service	TC-NFR-SEC-001; Security test (TC), inspection	Baseline
NFR-SEC-002	OBJ-08	Deployment, console, service	TC-NFR-SEC-002; Security test (TC), inspection	Baseline
NFR-SEC-003	OBJ-08	Deployment, console, service	TC-NFR-SEC-003; Security test (TC), inspection	Baseline
NFR-SEC-004	OBJ-08	Deployment, console, service	TC-NFR-SEC-004; Security test (TC), inspection	Baseline
NFR-SEC-005	OBJ-08	Deployment, console, service	TC-NFR-SEC-005; Security test (TC), inspection	Baseline
NFR-SEC-006	OBJ-08	Deployment, console, service	TC-NFR-SEC-006; Security test (TC), inspection	Proposed
NFR-SEC-007	OBJ-08	Deployment, console, service	TC-NFR-SEC-007; Security test (TC), inspection	Proposed
NFR-SEC-008	OBJ-08	Deployment, console, service	TC-NFR-SEC-008; Security test (TC), inspection	Proposed
NFR-SEC-009	OBJ-08	Deployment, console, service	TC-NFR-SEC-009; Security test (TC), inspection	Proposed

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
NFR-SEC-010	OBJ-08	Deployment, console, service	TC-NFR-SEC-010; Security test (TC), inspection	Proposed
NFR-SEC-011	OBJ-08	Deployment, console, service	TC-NFR-SEC-011; Security test (TC), inspection	Proposed
NFR-PRV-001	OBJ-08	Whole system	TC-NFR-PRV-001; Inspection	Baseline
NFR-PRV-002	OBJ-08	Whole system	TC-NFR-PRV-002; Inspection	Baseline
NFR-PRV-003	OBJ-08	Whole system	TC-NFR-PRV-003; Inspection	Proposed
NFR-PRV-004	OBJ-08	Whole system	TC-NFR-PRV-004; Inspection	Proposed
NFR-USE-001	OBJ-05	gps-dashboard-ui	TC-NFR-USE-001; Usability test, accessibility audit	Baseline
NFR-USE-002	OBJ-05	gps-dashboard-ui	TC-NFR-USE-002; Usability test, accessibility audit	Baseline
NFR-USE-003	OBJ-05	gps-dashboard-ui	TC-NFR-USE-003; Usability test, accessibility audit	Proposed
NFR-USE-004	OBJ-05	gps-dashboard-ui	TC-NFR-USE-004; Usability test, accessibility audit	Proposed
NFR-MNT-001	OBJ-06	Build, monitoring/	TC-NFR-MNT-001; Inspection, CI	Baseline
NFR-MNT-002	OBJ-06	Build, monitoring/	TC-NFR-MNT-002; Inspection, CI	Baseline
NFR-MNT-003	OBJ-06	Build, monitoring/	TC-NFR-MNT-003; Inspection, CI	Baseline
NFR-MNT-004	OBJ-06	Build, monitoring/	TC-NFR-MNT-004; Inspection, CI	Proposed
NFR-MNT-005	OBJ-06	Build, monitoring/	TC-NFR-MNT-005; Inspection, CI	Proposed
NFR-CMP-001	OBJ-07	Deployment, API	TC-NFR-CMP-001; Demonstration	Baseline
NFR-CMP-002	OBJ-07	Deployment, API	TC-NFR-CMP-002; Demonstration	Proposed
NFR-CMP-003	OBJ-07	Deployment, API	TC-NFR-CMP-003; Demonstration	Baseline
REG-001	OBJ-08	Operations, legal	TC-REG-001; Compliance review	Proposed
REG-002	OBJ-08	Operations, legal	TC-REG-002; Compliance review	Proposed
REG-003	OBJ-08	Operations, legal	TC-REG-003; Compliance review	Proposed
REG-004	OBJ-08	Operations, legal	TC-REG-004; Compliance review	Future
REG-005	OBJ-08	Operations, legal	TC-REG-005; Compliance review	Proposed
REG-006	OBJ-08	Operations, legal	TC-REG-006; Compliance review	Baseline
IFR-UI-001	OBJ-06	gps-dashboard-ui	TC-IFR-UI-001; Manual (TC)	Baseline
IFR-UI-002	OBJ-06	gps-dashboard-ui	TC-IFR-UI-002; Manual (TC)	Proposed
IFR-API-001	OBJ-06	REST controllers	TC-IFR-API-001; API tests (TC)	Proposed
IFR-API-002	OBJ-06	REST controllers	TC-IFR-API-002; API tests (TC)	Baseline
IFR-API-003	OBJ-06	REST controllers	TC-IFR-API-003; API tests (TC)	Baseline
IFR-API-004	OBJ-06	REST controllers	TC-IFR-API-004; API tests (TC)	Baseline
IFR-API-005	OBJ-06	REST controllers	TC-IFR-API-005; API tests (TC)	Proposed
IFR-KAF-001	OBJ-07	gps-configuration message/, KafkaTopicDeclarations	TC-IFR-KAF-001; GpsMessageJsonTest, StreamsNormalizationTest	Baseline
IFR-KAF-002	OBJ-07	gps-configuration message/, KafkaTopicDeclarations	TC-IFR-KAF-002; GpsMessageJsonTest, StreamsNormalizationTest	Baseline
IFR-KAF-003	OBJ-07	gps-configuration message/, KafkaTopicDeclarations	TC-IFR-KAF-003; GpsMessageJsonTest, StreamsNormalizationTest	Proposed
IFR-RDS-001	OBJ-06	redis/, RedisKeyNames	TC-IFR-RDS-001; RedisTargetTest	Baseline
IFR-RDS-002	OBJ-06	redis/, RedisKeyNames	TC-IFR-RDS-002; RedisTargetTest	Baseline
DR-001	OBJ-02	Database gps_tracking	TC-DR-001; Schema inspection, integration tests	Baseline
DR-002	OBJ-02	Database gps_tracking	TC-DR-002; Schema inspection, integration tests	Baseline
DR-003	OBJ-02	Database gps_tracking	TC-DR-003; Schema inspection, integration tests	Baseline

Requirement	Objective	Design component	Verification (test case; existing tests)	Status
DR-004	OBJ-02	Database gps_tracking	TC-DR-004; Schema inspection, integration tests	Baseline
DR-005	OBJ-02	Database gps_tracking	TC-DR-005; Schema inspection, integration tests	Baseline
DR-006	OBJ-02	Database gps_tracking	TC-DR-006; Schema inspection, integration tests	Baseline
DR-007	OBJ-02	Database gps_tracking	TC-DR-007; Schema inspection, integration tests	Baseline
DR-008	OBJ-02	Database gps_tracking	TC-DR-008; Schema inspection, integration tests	Proposed
DR-009	OBJ-02	Database gps_tracking	TC-DR-009; Schema inspection, integration tests	Baseline
DR-010	OBJ-02	Database gps_tracking	TC-DR-010; Schema inspection, integration tests	Baseline

Requirement count: 283 (Baseline 243, Proposed 33, Future 7).

8.3 Open Issues and Decisions Register

Table 74: *Open issues (TBD)*

ID	Issue	Owner	Needed by
OI-01	Confirm the performance and availability thresholds (NFR-PERF-001 to 005, NFR-AVL-001) after a baseline test on production-like hardware.	Technical lead	Before baseline 1.0
OI-02	Define user roles and permissions for multi-user access (FR-AUTH-010).	Product owner	Before external users
OI-03	Decide how the service API is protected: network isolation or service token (NFR-SEC-007).	Security reviewer	Before production
OI-04	Define retention per data type and tenant, and the deletion job (NFR-PRV-003, REG-001).	Product owner, legal	Before production
OI-05	Legal confirmation of REG-001 to REG-005 applicability and obligations.	Legal counsel	Before production
OI-06	API versioning scheme (/api/v1) and OpenAPI publication (IFR-API-005).	Technical lead	Release 0.2
OI-07	Repair job for trips with late readings (FR-TRP-011): trigger, schedule and audit.	Technical lead	Release 0.3
OI-08	Device protocols to support first in the device parser (FR-ING-008, REG-004).	Product owner	Release 0.3

9 Appendices

9.1 Appendix A: Configuration Parameters

Defaults from application.yml. Requirements refer to these keys; changing a key changes the verified value accordingly.

Table 75: *Configuration parameters*

Key	Default	Used by
gps.processing-flow	kafka	FR-ING-007 (kafka | direct)
gps.distance.jitter-meters / jitter-max-speed-kmph	10 / 5	FR-NRM-005
gps.distance.max-implied-speed-kmph / min-jump-meters	200 / 500	FR-NRM-006
gps.trips.min-distance-meters / min-duration-seconds / stop-gap-seconds	200 / 180 / 3600	FR-TRP-001 to 003
gps.stream.history-length	1000	FR-VST-004
gps.stream.stoppage.radius-meters / min-duration-seconds / stationary-speed-kmph	100 / 180 / 3	FR-STP-001
gps.stream.stoppage.exit-confirm-readings / max-gap-seconds / merge-gap-seconds	2 / 1800 / 180	FR-STP-002
gps.stream.ignition.virtual-off-after-seconds / idle-speed-kmph / idle-alert-seconds / virtual-debounce-readings	300 / 3 / 300 / 2	FR-IGN-001, FR-IGN-003
gps.stream.vehicle-state.gps-sigma-meters / acceleration-sigma	12 / 1.5	FR-VST-002
gps.stream.enrichment.weather-enabled / traffic-radius-meters / timeout-millis	true / 1000 / 3000	FR-STP-004, FR-STP-005
gps.kafka.retention / replication-factor / min-insync-replicas	31d / 1 / 1	NFR-CAP-002, NFR-REL-004
gps.journal.directory / records-per-segment / fsync-every-record / drain-batch-size	./data/ingest-journal / 1000 / true / 5000	FR-ING-002 to 005
gps.streams.enabled / normalization / threads / standby-replicas	true / owner / 3 / 0	FR-KS-001, FR-KS-002
gps.streams.max-lateness / seen-readings-retention	24h / 24h	FR-NRM-001, FR-NRM-002
gps.streams.window-size / window-grace / window-retention / parity-grace	5m / 2m / 6h / 2m	FR-KS-002, FR-KS-003
gps.streams.normalizer-guarantee / analytics-guarantee	exactly_once_v2 / at_least_once	FR-KS-001
gps.reports.zone / refresh-every / max-gap-seconds / max-stopped-gap-seconds / backfill-days	Asia/Kolkata / 5m / 600 / 3600 / 62	FR-RPT-001 to 003
gps.admin.reset-enabled	true	FR-ADM-003, NFR-SEC-010
gps.simulator.vehicle-count / pings-per-vehicle / minimum-pings-per-trip	150 / 500 / 200	FR-SIM-001
gps.simulator.timing / catch-up-minutes / catch-up-speed-factor	faithful / 120 / 30	FR-SIM-002
gps.simulator.devices-with-ignition / devices-with-odometer / idling-devices	0.5 / 0.5 / 0.25	FR-SIM-004
gps.connections.redis-command-timeout / redis-connect-timeout / http-timeout	5s / 3s / 5s	FR-CON-006
Speed rule default (Redis)	60 km/h, 3 pings, 60 s, hysteresis 5, tolerance max(10, 15 %), gap 180 s	FR-SPD-001 to 005

Key	Default	Used by
Fuel assumptions default (report_setting)	diesel, ₹92/L, 4 km/L, 2.5 L/h idle, 60 km/h, 20 %, 2.68 kg CO_2/L	FR-RPT-008
Console session (dashboard env)	12 h; 30 days remembered; 5 failures → 60 s lock	FR-AUTH-005, FR-AUTH-006

9.2 Appendix B: Requirement Statement Checklist

Every requirement in this document was checked against the following questions, taken from the companion SRS Rules and Regulations [R3]:

- Is it one obligation, stated with “shall”, and free of vague words (fast, user-friendly, real-time, secure) unless they are measured?
- Can it be verified by the stated method, and is the expected result unambiguous?
- Does it have a unique identifier, a priority, a status and a trace to an objective?
- Are the invalid-input, duplicate, late, missing-permission and dependency-failure cases covered for its module?
- Does it agree with every other requirement, figure and table in this document?